

# Vaccination and Economic Recovery<sup>\*</sup>

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## Abstract

The COVID-19 pandemic severely hampered human mobility and disrupted a series of economic activities. However, the role of COVID vaccination in the subsequent economic recovery remains under-studied. I construct a novel shift–share instrument that leverages the strong correlation between vaccination rates and demographic composition by sex and age group to address endogeneity concerns. Using high frequency data, I find strong evidence that higher vaccine coverage increased mobility, reflected in more trips taken and a lower likelihood of staying at home. I then assess the effects of vaccination on multiple economic indicators. I find modest evidence that higher vaccine coverage increased labor market participation and employment rate, while the effects on new business applications and home mortgage applications are limited. The results suggest that vaccination made a positive contribution to the overall economic recovery.

**Keywords:** vaccination, mobility

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# 1 Introduction

The COVID-19 pandemic has become the most broad-sweeping public health crisis in decades (Beach et al., 2022). The following lockdowns aiming at containing the spread led to a sharp decline of human mobility (Alexander and Karger, 2023; Arambepola et al., 2023; Bonaccorsi et al., 2020; Caselli et al., 2022). COVID-19 vaccines were introduced in December 2020, after the initial swift recovery periods of the pandemic had ended (Wachter, 2021). By Spring 2022, the labor market outcome, as measured in unemployment rate, had largely recovered to the pre-pandemic level, although labor force participation remained lower compared to the pre-pandemic level (Forsythe et al., 2022). The recovery was also uneven across social demographic groups. For example, Pinka and Yue (2023) document that the recovery in employment among young adults aged 20-24 lagged behind those of 15-19-year-olds.

COVID lockdowns, which were introduced at the onset of COVID pandemic, sharply reduced mobility (Alexander and Karger, 2023; Arambepola et al., 2023; Bonaccorsi et al., 2020; Bounie et al., 2023; Caselli et al., 2022). These effects were evident in a decrease of unnecessary public visits (Alexander and Karger, 2023), and a rise of work-from-home (Alipour et al., 2021). By measuring the number of trips taken, mobility serves as a credible indicator of the post-pandemic economic recovery. Accompanied by the decline in mobility, evidence suggests that small business revenue and consumer spending have declined since the onset of the lockdown (Alexander and Karger, 2023; Bounie et al., 2023). Consistent with this, Cepparulo (2025) finds that mobility, especially in retail and recreational spaces, is associated with higher in-store retail spending. On the other hand, mobility ran afoul of a range of policies that aimed at reducing COVID spread by limiting social contact (Hale et al., 2021). An increase in urban mobility can potentially increase COVID spread, mitigating the effects of COVID containment measures (Camehl and Rieth, 2023; Coven et al., 2023; Fang et al., 2020; Glaeser et al., 2022). Given these economic implications of mobility, this study first aims to bridge the gap examining the impacts of COVID vaccine coverage on post-pandemic recovery. Moreover, I delve into the economic implications by examining a variety of economic outcomes, such as the labor

market participation rate, employment rate, new business applications, and home mortgage applications.

To address the concern of endogeneity, I propose a novel “Bartik-style” shift-share instrument for vaccine coverage, in the sense that I exploit the strong variations in vaccine uptake across different age and sex groups that come through pre-determined age and sex composition. I motivate this strategy by addressing three potential sources of endogeneity that an OLS model faces. To the best of the author’s knowledge, this is the first article that uses a shift-share instrument to identify the causal effects of vaccination on health outcomes, in a way that is plausibly not subject to the “collective invalidation” concern raised by [Gallen \(2020\)](#).

Using high-frequency daily data on vaccination from the CDC and Daily Mobility Statistics from the Bureau of Transportation Statistics (BTS), my results suggest that COVID vaccine coverage, defined as the proportion of the population that received at least one shot of COVID vaccine, increased mobility, reflected in an increase in trips and a decrease in the stay-at-home population. My baseline IV estimates suggest that a one percent increase in first-dose vaccination rate is associated with 0.029 increase in daily number of trips taken per person and a 0.162 percent decrease in the stay-at-home population. I then address multiple potential threats to identification, including concerns on endogeneity and measurement errors. These results are further confirmed using Google Mobility Data as an alternative measure of mobility.

I then turn to examining the broad economic impacts of COVID vaccine coverage on 4 indicators: labor force participation rate, employment rate, new business applications per 1,000 population, and home mortgage applications per 1,000 population. Given the constraints of the data, the analysis is conducted at a state-by-month level. My results provide modest evidence of a positive impact towards the labor market participation rate and the employment rate, while the evidence on new business applications and home mortgage applications is limited.

My study makes multiple contribution to the literature. First, my study directly contributes to the literature that examines the effects of COVID vaccination on economic and health outcomes ([Agrawal et al., 2021](#); [Aslim et al., 2024, 2025](#); [Barro, 2022](#); [Chaudhuri and Howley, 2022](#);

Freedman et al., 2025; Hansen and Mano, 2022). Second, I contribute to the literature that examines the interaction of COVID spread and mobility (Camehl and Rieth, 2023) as well as on the ‘moral hazard’ effects of vaccination (Agrawal et al., 2022; Coven et al., 2023; Fang et al., 2020; Glaeser et al., 2022). Third, while not the main interests of my study, I broadly contribute to the literature that examines the social/policy determinants of vaccination (Abel et al., 2025; Andor et al., 2025; Červený et al., 2025; Dai et al., 2021; D’Souza et al., 2025; Garrouste et al., 2025; Klaesson et al., 2025).

## 2 Literature Review

### 2.1 Health effects of COVID vaccination

COVID vaccine was shown to effectively save lives by reducing the risk and severity of COVID infection in both economics literature (see Agostini et al. (2024); Freedman et al. (2025)) and health literature (see Gupta et al. (2021); Watson et al. (2022)). The immunity to COVID-19 infection offered by vaccines has also been shown to reduce mortality (Sandmann et al., 2021). In a global context, Agrawal et al. (2023) find that COVID vaccination reduced all-cause mortality, saving an estimated 2.4 million lives. Using a double-debiased machine learning model, Girma and Paton (2024) find mild evidence that COVID-19 vaccine uptake reduced mortality in elder-care facilities in Italy. Furthermore, Barro (2022) finds that COVID vaccination reduced mortality not only from COVID but also of other causes. While he also suggests that the effects on mortality weakened after the emergence of new variants.

Previous economics literature had shown that vaccination had a wide range of impact on people’s health. Utilizing the Household Pulse Survey data, Agrawal et al. (2021) find that COVID vaccination led to improved mental health. In particular, they observe stronger effects among economically disadvantaged individuals. Using an UK survey data, Chaudhuri and Howley (2022) find positive effects of COVID vaccination on mental well-being, with stronger effects among the elderly and clinically vulnerable individuals.



Aslim et al. (2024) find that COVID vaccination is associated with a lower likelihood of skipping care, as a result of the alleviated concerns of COVID infection. Utilizing both instrumental variable (IV) and regression discontinuity design (RDD) methods, Aslim et al. (2025) find consistent evidence that COVID vaccination reduced food insecurity and the reliance on Supplemental Nutrition Assistance Program (SNAP). They attribute the improvement in food security to increased optimism on future employment and income.

By reducing the risk and severity associated with COVID infection, vaccination could theoretically alter individuals' health-related behaviors (see Ehrlich and Becker (1972); Peltzman (1975); Kremer (1996)). Previous research had investigated the effects of COVID vaccination on individuals' engagement in risky behaviors. As a US study, Agrawal et al. (2022) find that vaccination did not reduce risk mitigation behaviors such as wearing mask or washing hand. Similarly, Hwang et al. (2024) find little evidence of risk compensating behaviors following COVID vaccination in South Korea.

## 2.2 Determinants of COVID vaccination

A body of literature has examined the determinants of the likelihood to get vaccinated. In addition to factors primarily affecting vaccine supply, factors like age, education, income, and the severity of the pandemic influences the decisions to get vaccinated (Deb et al., 2023; Kafadar et al., 2023; Randelović and Tanasković, 2024). Additionally, views towards vaccines (vaccine hesitancy) (Gans, 2023; Rawlings et al., 2022) and peer effects/herd behaviors (Agranov et al., 2021; Epstein et al., 2025) played an important role on vaccination decisions.

Furthermore, previous research find that places within proximity to the vaccine administration sites (Brownstein et al., 2022) and regions with policies like the vaccine lotteries (Brehm et al., 2022), vaccine certificates (Oliu-Barton et al., 2022), and vaccine mandates (Nguyen et al., 2025) are associated with increased vaccination rates. While such policies were intended to boost vaccine uptake, they are likely correlated with other interventions, such as eviction moratorium (An et al., 2022), that may have a direct impact on economic activities. It is important

to address the associated endogeneity (see Section 3 for empirical analysis addressing these endogeneity).

The effectiveness of such policy interventions aiming at promoting vaccine uptake remains debated. Exploiting variations in university vaccine mandates, [Acton et al. \(2024\)](#) find a notable reduction of COVID cases and deaths in surrounding areas. On the other hand, [Abouk et al. \(2025\)](#) find that vaccine mandates for healthcare workers reduced employment in the healthcare industry, potentially worsening the existing shortage of healthcare staff.

## 2.3 Effects of COVID Vaccination on Economic Activity

As a cross-country study, [Deb et al. \(2022\)](#) examine the effects of “vaccine surprises,” measured as the difference between the number of predicted vaccine supply and the vaccine uptakes, on proxies for economic activities, including  $NO_2$  and  $CO$  emissions and mobility. They find that COVID-19 vaccination contributes to significant increases in such activities.

[Hansen and Mano \(2022\)](#) is the study that is the most closely related to this one. In which, the authors assess the effects of initial COVID-19 vaccine uptake on economic activities including credit card spending, mobility at the workplace, and unemployment insurance claims. To isolate the casual effects, they construct an instrument in spirit of Bartik using the interaction of pharmacy density and state-level vaccine supply. They find that vaccination had generally positive effects, including an increase in credit card spending and workplace mobility, as well as a reduction in unemployment insurance claims.

This study makes multiple contributions to [Hansen and Mano \(2022\)](#). First, I expand the observation windows to June 2022, whereas the their study ended in July 2021, which likely did not capture the full dynamic effects. Second, they construct a Bartik-like instrument using the density of pharmacies by county as the share and state-level vaccine allocation as the shocks. However, this design likely weakens the instrument’s ability to capture county-level variation over time. I construct a shift-share instrument using county-level demographic composition on age and sex, which alleviates the concerns that the pre-determined pharmacy locations may

correlated with un-observables. Third, using data from the Bureau of Transportation Statistics, I estimate the effects on the number of trips taken per person and the percentage of individuals staying-at-home, which provide clearer economic interpretation. Fourth, I examine the effects of COVID-19 vaccination on additional economic indicators, thereby providing insights to broader economic implications.

## 3 Empirical Strategy

### 3.1 Data Selection

**CDC Vaccination Data** My primary explanatory variable is the cumulated daily first dose vaccination administration rate obtained from the CDC.<sup>1</sup> This dataset records the number of individuals that received at least one dose of COVID vaccine. In other words, our explanatory variable is the cumulative first dose vaccination rate by county. This data starts on December 13, 2020, when the first recorded dose of COVID vaccine was administered and ends in May 10, 2023 (see Figure A2 for the start and end dates of vaccination data and the corresponding COVID timeline). The data was reported on a daily basis from the beginning of the series until June 6, 2022, after which it was reported weekly.

The vaccination data was available December 13, 2020 to May 10, 2023, with data provided at a daily level from December 13, 2020 to June 16, 2022, and at a monthly level from June 16, 2022 to May 10, 2023. For consistency, I select December 13, 2020 to June 16, 2022 as my main estimation periods.

To construct a shift-share IV (see Section 3.4), I obtain national level vaccination data that covers the first-dose vaccine administration rates across different age and sex groups from the CDC.<sup>2</sup> The advantage of this dataset is that, while it is not available at the county level, it

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<sup>1</sup>The dataset is named “COVID-19 Vaccinations in the United States, County.” See the archived CDC COVID data at [https://covid.cdc.gov/covid-data-tracker/?utm\\_source=chatgpt.com#archived](https://covid.cdc.gov/covid-data-tracker/?utm_source=chatgpt.com#archived).

<sup>2</sup>The dataset is named “COVID-19 Vaccination Age and Sex Trends in the United States, National and Jurisdictional”. See the archived CDC COVID data at [https://covid.cdc.gov/covid-data-tracker/?utm\\_source=chatgpt.com#archived](https://covid.cdc.gov/covid-data-tracker/?utm_source=chatgpt.com#archived).

has more detailed information on the vaccination rates by age and sex, which allows me to construct the “shocks” in a shift-share IV. Furthermore, I obtain the age and sex composition at a county-level from the Census. This allows me to construct the initial “share” of the shift-share IV.<sup>3</sup>

One potential concern of the CDC vaccination data is that it may over count the first doses. Because the follow-up vaccine doses are not perfectly linked to the same receivers, the first dose vaccine uptake is likely over-counted. This would be concerning if there exists systematic patterns regarding the accuracy of this measurement. I attempt to address this issue by utilizing a predicted first-dose vaccine uptake rate (detailed summary see Appendix A.2).

**BTS Daily Mobility Statistics** I obtain the Daily Mobility Statistics from the Bureau of Transportation Statistics (BTS), which sources mobile location data from multiple data vendors and aggregate at the county level (see [U.S. Department of Transportation, Bureau of Transportation Statistics \(2023\)](#)).<sup>4</sup> A unique advantage of the dataset has a higher cellphone penetration rate compared to other data sources ([U.S. Department of Transportation, Bureau of Transportation Statistics, 2023](#)). Although the data is not the most granular at geographic level, finer spatial details are not required for the purpose of this study. My primary outcome variables include two measures of mobility: (1) the number of trips per capita, (2) the percentage of population staying-at-home. As defined by the BTS, the total number of trips includes all movements lasting at least 10 minutes outside an individual’s home. The advantage of this data is that the magnitude of the estimates has a clear economic interpretation, which corresponds to changes in the average number of trips taken per person and in the share of the population staying at home.

**Google Mobility Data** An alternative measure for mobility is from the Google COVID-19 Com-

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<sup>3</sup>After matching the CDC data to the Census, I obtain the number of individuals who have received at least one dose of the vaccine in the following age groups by sex: 0–4 years, 5–17 years, 18–24 years, 25–39 years, 40–49 years, 50–64 years, 65–74 years, and over 75 years.

<sup>4</sup>See related studies that used GPS location data—[Couture et al. \(2022\)](#); [Coven et al. \(2023\)](#); [Glaeser et al. \(2022\)](#).

munity Mobility Report, where it recorded daily mobility patterns during the COVID-19 pandemic until October 2022 (Google, 2020). The data presents how the daily mobility in six categories of places, including “Retail & recreation,” “Grocery & pharmacy,” “Parks,” “Transit stations,” “Workplaces,” “Residential,” changed compared to the pre-pandemic level.<sup>5</sup> I obtain daily state-level data as the supplemental outcomes for my analysis. The estimation strictly follows my main analysis. There are two benefits using this alternative data. Firstly, as the data provides mobility in different areas, it gives confidence that my baseline results are not driven by confounding trips to get vaccinated. Secondly, it allows me to provide extra insight on the heterogeneous impact of COVID vaccination on different trip types.

The summary statistics is presented in Appendix Table A1. A person undertakes an average of 3.9 trips per day during the sample periods. On average, 18.4 % of the population stays at home on any given day. Additionally, as expected, the vaccination rate increases with age.

### 3.2 OLS model

I first estimate the effects of vaccination rates on mobility, measured as the number of trips per capita and the percentage of the population staying at home. The estimation equation can be stated as follows,

$$y_{it} = \beta \times Vac_{i,t-1} + \mathbf{X}'_{it} + \gamma_i + \delta_t + \epsilon_{it} \quad (1)$$

where  $y_{it}$  denotes the outcome variable,  $Vac_{i,t-1}$  denotes the vaccine coverage,  $\mathbf{X}'_{it}$  represents a vector of covariates<sup>6</sup> and their corresponding coefficients,  $\gamma_i$  captures a county fixed-effects,  $\delta_t$  captures a day fixed effects, and  $\epsilon_{it}$  is the error term.  $\beta$  captures the effects of COVID vaccine

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<sup>5</sup>These could be further categorized as public spaces including “Retail & recreation,” “Grocery & pharmacy,” “Parks,” “Transit stations,” “Workplaces,” and residential places such as “Residential,” where the indicators for the first category were measured as changes in total visits to each location, and the second category as changes in the duration of stay in residential areas.

<sup>6</sup>The covariates include ethnicity, educational attainment, and poverty rate, obtained from the U.S. Census Bureau using the “tidycensus” package in R. The data is based on county-level ACS 5-year estimates, with the end year corresponding to the current year.

coverage on outcomes.

I also lag the explanatory variable,  $Vac_{i,t-1}$ , by one period. Intuitively, because getting vaccinated would require travel, the act of traveling itself could serve as a confounding factor. I lag the explanatory variable so that vaccine coverage is used to explain future mobility. Additionally, I estimate the effects of COVID vaccination on trips of different distance (see detailed summary in Appendix A.3).

### 3.3 Threats to identification

**Confounding Health Policies** The state and federal governments implemented various policies to reduce mobility, aiming to limit COVID spread (Goolsbee and Syverson, 2021; Hale et al., 2021). A potential threat to my identification is that local vaccination rate correlates with other factors that affect the outcomes (see factors that affected COVID severity (Alsina-Pujols and Vall Castelló, 2025)). For example, stringent lockdown policies that aimed to reduce mobility (Hale et al., 2021) were likely correlated with vaccine mandates that aimed to boost vaccine uptake (Abouk et al., 2025; Nillasithanukroh et al., 2025). There were likely various such policies that I cannot simply observe. If this is the case, the OLS estimates are biased due to endogeneity. The instrumental variable approach can effectively isolate the casual effects of vaccination on my outcomes from state-time varying confounders (e.g. state-specific shocks like stay-at-home orders).

**Reverse Causality** In this context, reverse causality implies that the outcomes of our interests, geographic mobility, had an impact on my explanatory variable, vaccination rate. d’Albis et al. (2024) find empirical evidence that lagged increases in geographic mobility led to increases in COVID hospitalizations. At the same time, they argued that geographic mobility and COVID infections are “jointly determined.” Additionally, research show that geographic mobility is associated with higher COVID infection rates (Camehl and Rieth, 2023; Coven et al., 2023; Fang et al., 2020; Glaeser et al., 2022). The perception of the elevated infection risks could reversely

impact individuals' decisions whether to get vaccinated or not (see related research on the willingness to vaccinate (Coe et al., 2022; Karlsson et al., 2021; Qiao et al., 2022)). Indeed, Oster (2018) find that pertussis outbreak induced a greater pertussis vaccine uptake.

**Unobserved Social Determinants** An additional threat to identification is the unobserved social determinants of vaccine uptake that might correlate with outcomes. An instrumental variable approach is likely preferred as long as the instrument is uncorrelated with the potential confounders. As Adão et al. (2019) illustrated, the shares or shifts are required to be “as good as random” conditional on the other covariates. Given this assumption, I can construct a plausibly exogenous instrument while addressing those potential violations.

### 3.4 Instrumental variable approach

As illustrated, the explanatory variable is likely correlated with un-observables that open “back door” pathway to the outcome. An instrumental variable method is commonly used to isolate the causal effects from the endogenous explanatory variable to the outcome.

Previous economics research has adopted multiple instruments to isolate the causal effects on vaccination on various outcomes. For example, Aslim et al. (2024, 2025); Freedman et al. (2025) used the gradual roll-out of vaccine by age and state as an instrument. Barro (2022), on the other hand, used the 2020 presidential election vote share as an instrument for vaccination rate. There are three challenges applying their approach to this study. Firstly, their research investigated the effects of vaccination at the individual level, whereas my analysis cannot, since I lack data on individual uptake. Secondly, the time-window of the gradual roll-out of vaccine ranges from 2020 December to 2021 May, which leaves a relative short periods for the study. Thirdly, it remains unconvincing to use partisan vote as an instrument for vaccine uptake, as it may potentially correlate with other unobservable. Instead, I adopt a Bartik shift-share instrumental variable (SSIV) approach (Bartik, 1991), in which the instrument is constructed by interacting a pre-determined share and a common shock.



A shift-share IV approach is likely preferred to address the endogenous issues mentioned above. Firstly, a shift-share IV uses the predicted value of the explanatory variable, which can adequately remove the regional confounding policies. Secondly, a shift-share IV builds on pre-determined “shares,” which are unlikely affected by reverse causation. Thirdly, a shift-share IV relies on either the exogeneity of “shares” or “shifts,” which plausibly alleviates concerns on un-observed social determinants.

SSIV contains two parts: (1) a pre-determined weights that determines the level of exposure to shocks, and (2) an overall level of shocks through pre-determinants weights on sub-divisions. The shift-shared instrument is widely used in studying the economic effects of immigration (Card, 2001) and trade (Autor et al., 2013).<sup>7</sup> It is not uncommon to use social demographic composition as the sub-level for an SSIV. For instance, Mohnen (2025) has used an SSIV, exploiting variation in employment rates by age among individuals over 45, to examine the effects of senior retirement on youth employment.

SSIV is also widely used in urban economics literature. Gilraine and Zheng (2024) decomposes different pollution sources to examine the effects of air pollution to students’ performance. Boustan et al. (2024) exploits the country-of-origin variation to assess the effects of Asian immigration on White students’ enrollment in California. Komarek et al. (2022) leverage the sub-level variations of government procurement spending across industries to examine effects of Budget Control Act of 2011 on local labor market outcomes. Mocanu and Tremacoldi-Rossi (2023) draw on the country-of-origin variation to estimate the effects of international students on house prices and rents. Wang et al. (2025) decompose product categories to examine the effects of export on local land supply.

Intuitively, I exploit the strong variation of vaccine uptake across age and sex groups, along

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<sup>7</sup>For instance, previous research has used the decomposition of national origins to examine the effects of immigration/emigration on local labor supplies (Cortés and Tessada, 2011), natives’ health (Giuntella and Mazzonna, 2015), fertility (Romiti, 2017), natives’ work schedule (Bond et al., 2023), FDI (Burchardi et al., 2019), patents and economic growth (Burchardi et al., 2020; Hunt and Gauthier-Loiselle, 2010), and entrepreneurship (Anelli et al., 2023). Additionally, an emerging stream of literature exploits the shares of employment by industry to examine the impact of robotization on productivity (Acemoglu et al., 2020), employment and wage (Acemoglu and Restrepo, 2020), workers’ safety (Gihleb et al., 2022), college enrollment (Di Giacomo and Lerch, 2023), fertility rate (Anelli et al., 2024), and various labor market outcomes in China (Giuntella et al., 2024).



with the pre-pandemic age and sex makeup of each county, to construct an expected rates of vaccination. Specifically, I use the proportion of age group by sex in 2019 as the “shares” of the instrument.<sup>8</sup> I use the national level vaccination rates as the “shifts” of the instrument. The SSIV is stated as,

$$\widetilde{Z}_{it} = \sum_k \underbrace{\frac{demgroup_{ik}^{2019}}{pop_i^{2019}}}_{shares} \times \underbrace{VAC_{kt}}_{shifts} \quad (2)$$

where  $\widetilde{Z}_{it}$  denotes the instrumental variable varied by county  $i$ , and date  $t$ . I use the 2019 demographic makeup,  $\frac{demgroup_{ik}^{2019}}{pop_i^{2019}}$ , as the “shares” of the instrument and national vaccination rate adjusted by 2019 population,  $VAC_{kt}$ , as the “shifts” of the instrument. The model used for estimation could be stated as,

$$y_{it} = \beta \times \widehat{VAC}_{i,t-1} + \mathbf{X}'_{it} + \gamma_i + \delta_t + \epsilon_{it} \quad (3)$$

where all notations follow Equation (1).  $\widehat{VAC}_{i,t-1}$  is obtained from the first-stage regression,

$$VAC_{i,t-1} = \alpha_1 \widetilde{Z}_{i,t-1} + \mathbf{X}'_{i,t-1} + \theta_i + \iota_{t-1} + v_{i,t-1} \quad (4)$$

where  $VAC_{i,t-1}$  is the endogenous vaccination rate,  $\widetilde{Z}_{i,t-1}$  is the predicted vaccination rate,  $\theta_i$  and  $\iota_{t-1}$  are county and day fixed effects respectively, and  $v_{i,t-1}$  is the error term.

### 3.5 Instrument validity

For the instrument to be valid, two assumptions must be satisfied: (1) the SSIV must be strongly correlated with the explanatory variable, (2) the SSIV must only affect the outcomes through the explanatory variable.

The first assumption, relevance assumption, is easily testable. I show that this assumption

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<sup>8</sup>These include 8 age groups including: 0–4 years, 5–17 years, 18–24 years, 25–39 years, 40–49 years, 50–64 years, 65–74 years, and over 75 years, and 2 age groups by male and female, which summed to 16 groups in total.

is valid from my first-stage results.

For the second assumption to hold, it must be the case that, conditional on controls either the shares are exogenous or there are many exogenous shifts (Borusyak et al., 2022, 2025; Goldsmith-Pinkham et al., 2020). There are two reasons that the second case, assuming exogenous shifts, is unlikely to hold. Firstly, the timing of vaccine eligibility for different age groups is staggered, which makes the vaccination rates across age groups not random. Secondly, different age groups ultimately face different risks and perceptions towards COVID, which in turn makes vaccination rates across age group not random.

My identification relies on a “share based” approach (Borusyak et al., 2022, 2025; Goldsmith-Pinkham et al., 2020). In other words, I assume that the age and sex composition in each county pre-pandemic is “as good as” randomly assigned. Since the identification assumption relies on the exogeneity of the shares (the pre-determined demographic composition by gender and age group), I follow by discussing potential threats to this assumption and provide additional evidence on the instrument validity.

The broad idea of my shift-share IV is to exploit the strong variation in vaccine uptake among different demographic (age and sex) groups, which comes through the pre-determined county-level age and sex makeup. The exogeneity assumption would be violated if different demographic groups exhibit systematically different patterns in mobility. That is, aside from age and sex being strong determinants for vaccination, any relationship between age and sex and mobility would violate the exogeneity assumption. Indeed, previous research show that trip participation and frequency varies by different demographic group (Bartl et al., 2024), and that commuting distance also differs among these groups (Manaugh et al., 2010). Manaugh et al. (2010) show that commuting distance is positively correlated with male individuals and negatively correlated with age. However, the biases arising from this are likely small, as age and sex appear to be clearly stronger determinants of vaccination. Including unit-level controls likely mitigated the concern of un-wanted direct relationship between shares and mobility. To further address this concern, I estimate the relationship between vaccine coverage and

the *changes* in daily mobility, which is arguably orthogonal to pre-determined shares (detailed analysis see Appendix A.6).

A second threat to my identification is that the population with different ages exhibit systematically different beliefs about vaccination (Barro, 2022; Cao et al., 2022). The relationship was found to be determined by the county's political leanings (Albrecht, 2022). This is concerning, because counties with different political leanings may vary in more than just vaccine uptake, they may also differ in factors that directly related to mobility. Additionally, there is little time variation in political leanings within a county, leaving it not feasible to control for. I alleviate this concern by looking into counties with similar political leanings using the quantile of Democratic/Republican presidential election vote shares (detailed analysis see Appendix A.5).

The impact of COVID-19 on health, measured as the severity of infection, varies by age group (Alsan et al., 2021), with the older population generally exhibiting more severe symptoms compared to the younger population. As a result, the impact of COVID-19 on health is likely greater in areas with older population. The elevated risks associated with older population might compromise their willingness to undertake trips (see (Deiana et al., 2025)). At the same time, the older population are more likely to get vaccinated compared to the younger vaccination. Thus, my SSIV, which is the predicted vaccination rate, will be positively correlated with the severity of COVID-19. However, I argue that this is unlikely to introduce endogeneity because the time periods for my main analysis started from 2020 December, instead of the onset of the pandemic. The states with older or younger population already experienced a reduction of trips, though the effect sizes are different for states with different age makeups.<sup>9</sup> In this scenario, the severity of COVID-19 pandemic on individuals changes through vaccination, which the effects on mobility would thus be captured by my intended estimator. It remains possible that the severity of COVID-19, caused by new variants, varied across time, which may correlate with my instrument and directly affects the outcomes (Hoogenboom et al., 2021; Jassat et al.,

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<sup>9</sup>See a related European study—Caselli et al. (2021).

2022; Leung et al., 2020). I further adopt an exercise estimating within different time windows, which could plausibly alleviate the concerns about the “new variant shock” (detailed analysis see Appendix A.4).

Furthermore, following the empirical tests suggested by Goldsmith-Pinkham et al. (2020), I show evidence on the validity of this assumption with a pre-trends analysis (detailed analysis see Appendix A.7). I also probe the robustness of my analysis by using lagged sex and age distribution, or “lagged shares,” which are plausibly more exogenous, albeit at risk of reducing the predictive power of the first-stage (detailed analysis see Appendix A.8).

## 4 Results

I first present the effects of vaccine coverage on mobility. The upper panel of Table 1 presents the effects vaccination coverage on the number of trips taken per capita, and the bottom panel presents the effects of vaccination coverage on the percentage of population staying-at-home. I present the reduced form estimates in Column (1), the OLS estimates without and with covariates in Columns (2) and (3), and IV estimates in Columns (4) and (5).

In general, the estimates are consistent across both panels in terms of significance and signs. The F-statistics show sufficient strength for the first-stage regressions (first stage results see Appendix Table A9). The upper panel results suggest consistent evidence that vaccination coverage led to increased mobility, measured as the number of trips taken per person. The coefficients can be interpreted as follows: according to Column (5), a one percent increase in the vaccination coverage led to an average increase of 0.029 trips per person. The bottom panel results suggest that higher vaccination coverage reduced the likelihood of a person staying at home. The estimate in Column (5) suggests that, a one percent increase in the vaccination coverage reduced the population staying-at-home by 0.162 percent. Specifically, I find that the IV estimates are larger in magnitudes in both panels compared to the OLS estimates, suggesting that the unobserved factors likely cause a downward bias in my estimates.

I turn to present the results using Google Mobility data in Table 2, where Panels A and B present the OLS estimates without and with controls, and Panels C and D present the 2SLS estimates without and with controls. I find consistent results that COVID vaccine coverage increase mobility by increasing visits to public spaces from Columns (1) to (5). Also, the results suggest a decrease in the time stayed in residential areas according to Column (6). Turning into the 2SLS results, while the effects on visits to parks and transit stations lost significance, I still observe consistent signs suggesting an increase in visits to public spaces and a decrease of stay in residential areas. In summary, these results clearly rule out the possibility that the increase in mobility was driven by the potentially confounding factor of trips to get vaccinated.

## 5 Economic Activities

This chapter examine the effects of vaccination on general economic activities. I obtain four common macroeconomic indicators to assess the effects of vaccination on economic activities (similar research, see (Brodeur and Yousaf, 2025)). Especially, I examine the effects on 4 economic outcomes, including the labor force participation rate, employment rate, new business application rate, and home mortgage application rate. The labor force participation rate is obtained from the Bureau of Labor Statistics (BLS) at a state-by-month level. The employment rate is obtained from the Bureau of Labor Statistics (BLS) at a state-by-month level. The business application rate is obtained from the Census Bureau at a state-by-month level. The home mortgage application rate data is from the Home Mortgage Disclosure Act (HDMA) data compiled by Neil Bhutta.<sup>10</sup> I conduct the analysis using a state-by-month panel. The estimation strategy follows my baseline approach, but with states as the cross-sectional units (rather than counties) and months as the time units (rather than days).

I present the estimated effects on such economic indicators in Table 3. From top to bottom, the table display the effects of COVID vaccine coverage on the labor force participation rate, the

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<sup>10</sup><https://sites.google.com/site/neilbhutta/data?authuser=0>

employment rate, new business applications per 1,000 people, and home mortgage applications per 1,000 people. Across all four panels, I find that the first-stage F statistics at 14.9 and 15.3, with or without controls included, respectively. While the F statistics are much lower compared to the ones in my baseline analysis, I find them above the conventional rule of thumb levels ( $F = 10$ ). Additionally, given the sample size is smaller at the state-by-month level, the first stage appears sufficiently strong to support the analysis.

I present the effects of COVID vaccine coverage on the labor force participation rate in the upper panel. According to OLS estimates, I find some evidence that vaccine coverage increased the labor force participation rate, with a one percent increase in vaccine coverage associated with 0.026 to 0.025 percent increase in labor force participation rate.<sup>11</sup> Turning into IV estimate, I find statistically insignificant effects, with one percent increase in vaccine coverage associated with 0.021 ( $P = 0.150$ ) and 0.017 ( $P = 0.204$ ) percent increase in the labor force participation rate. Turning to the effects the second panel, I find that a one percent increase in vaccine coverage associated with a statistically significant 0.041 to 0.038 percent increase in the employment rate. However, the OLS estimates can be biased due to endogeneity. Using IV methods, I find that a one percent increase in vaccine coverage associated with a statistically insignificant 0.025 ( $P = 0.160$ ) to 0.022 ( $P = 0.184$ ) percent increase in the employment rate. Although the IV estimates on labor market participation rate and employment rate are statistically insignificant, the results could suggest that the models are under-powered given the small sample size ( $N = 918$ ). The signs of the coefficients are consistent, indicating that vaccine coverage likely had a positive impact on these two labor market indicators. These results align with findings of Richardson (2025), suggesting reduced health risks associated with higher vaccine coverage likely boosted workers' labor supply. The third and forth panels present the estimated effects of COVID vaccine coverage on business applications and home mortgage applications. Both OLS and IV estimates show little effect on these two economic indicators. It is unsurprising that vaccination had little "recovery" effects on these economic indicators, as business and

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<sup>11</sup>This implies that if every resident in a county were vaccinated, the county would experience a 2.6 to 2.5 percent increase in the labor force participation rate.

home mortgage applications did not experience an decline during the COVID-19 pandemic (see Figure A3).

In summary, I find consistent signs suggesting that vaccination had a modest positive impact on labor force participation rate and employment rate. On the other hand, I find that COVID vaccine coverage had little impact on business applications and mortgage applications.

## 6 Conclusion & Discussion

This study examines the causal relationship between COVID vaccine coverage and the subsequent economic recovery. To address the endogeneity in vaccine uptake, I construct a novel shift-share instrument, exploiting the variation in national vaccination trends by age and sex group. Specifically, I leverage the pre-determined age-sex compositions by county to predict vaccination rate based on the national vaccination trends. I find that increasing first-dose vaccine uptake by one percent increased mobility, with individuals taking about 0.029 more trips per person on average and 0.162 percentage points fewer people remaining at home. Furthermore, I examine the impacts of vaccine coverage on economic activities. I find modest and positive impacts on the labor force participation rate and employment rate, but no impacts on new business application rate and home mortgage application rate. These results suggest that the COVID vaccine served as a powerful tool for economic recovery.

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## 7 Tables

|                                      | (1)                                   | (2)                  | (3)                  | (4)                  | (5)                  |
|--------------------------------------|---------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                      | Trips per capita                      |                      |                      |                      |                      |
|                                      | reduced form                          | OLS                  |                      | IV                   |                      |
| SSIV                                 | 0.027**<br>(0.013)                    |                      |                      |                      |                      |
| Vac Rate                             |                                       | 0.005***<br>(0.001)  | 0.005***<br>(0.001)  |                      |                      |
| Fitted Vac Rate                      |                                       |                      |                      | 0.029**<br>(0.014)   | 0.029**<br>(0.014)   |
| Num.Obs.                             | 1 714 707                             | 1 664 179            | 1 664 179            | 1 664 179            | 1 664 179            |
| Std.Errors                           | by: fips                              | by: fips             | by: fips             | by: fips             | by: fips             |
| FE: date                             | X                                     | X                    | X                    | X                    | X                    |
| FE: fips                             | X                                     | X                    | X                    | X                    | X                    |
| Covariates:                          | X                                     |                      | X                    |                      | X                    |
| IV Wald F statistics:                |                                       |                      |                      | 76.2                 | 75.8                 |
| * p < 0.1, ** p < 0.05, *** p < 0.01 |                                       |                      |                      |                      |                      |
|                                      | (1)                                   | (2)                  | (3)                  | (4)                  | (5)                  |
|                                      | Percentage of Population Stay-at-home |                      |                      |                      |                      |
|                                      | reduced form                          | OLS                  |                      | IV                   |                      |
| SSIV                                 | -0.157***<br>(0.022)                  |                      |                      |                      |                      |
| Vac Rate                             |                                       | -0.026***<br>(0.003) | -0.026***<br>(0.003) |                      |                      |
| Fitted Vac Rate                      |                                       |                      |                      | -0.162***<br>(0.026) | -0.163***<br>(0.026) |
| Num.Obs.                             | 1 716 578                             | 1 665 914            | 1 665 914            | 1 665 914            | 1 665 914            |
| FE: date                             | X                                     | X                    | X                    | X                    | X                    |
| FE: county                           | X                                     | X                    | X                    | X                    | X                    |
| Covariates:                          | X                                     |                      | X                    |                      | X                    |
| IV Wald F statistics:                |                                       |                      |                      | 76.2                 | 75.8                 |
| * p < 0.1, ** p < 0.05, *** p < 0.01 |                                       |                      |                      |                      |                      |

**Table 1: Effects of Vaccination on Trips and the Proportion of Population Staying-at-home**

Note: This table presents the estimated effects of COVID vaccine coverage on the average number of trips taken per person and the percentage of the population staying-at-home. Column (1) present the reduced form estimates. Columns (2) and (3) present the OLS estimates. Columns (4) and (5) present the IV estimates. Covariates including race, education, and poverty rates are included in Columns (1), (3), and (5). County and day fixed effects are included in all columns. Standard errors clustered at the county level are included in the parentheses. The estimation period is from December 13, 2020 to June 16, 2022.



|                                       | (1)<br>Retail & Recreation | (2)<br>Grocery & Pharmacy | (3)<br>Parks        | (4)<br>Transit Stations | (5)<br>Workplaces   | (6)<br>Residential   |
|---------------------------------------|----------------------------|---------------------------|---------------------|-------------------------|---------------------|----------------------|
| <i>Panel A: OLS without Controls</i>  |                            |                           |                     |                         |                     |                      |
| Vac Rate                              | 0.109***<br>(0.009)        | 0.057***<br>(0.008)       | 0.460***<br>(0.038) | 0.063***<br>(0.019)     | 0.020***<br>(0.004) | −0.041***<br>(0.002) |
| <i>Panel B: OLS with Controls</i>     |                            |                           |                     |                         |                     |                      |
| Vac Rate                              | 0.110***<br>(0.009)        | 0.057***<br>(0.008)       | 0.458***<br>(0.038) | 0.063***<br>(0.019)     | 0.021***<br>(0.004) | −0.040***<br>(0.002) |
| <i>Panel C: 2SLS without Controls</i> |                            |                           |                     |                         |                     |                      |
| Fitted Vac Rate                       | 0.858***<br>(0.099)        | 0.505***<br>(0.072)       | 0.048<br>(0.286)    | 0.004<br>(0.122)        | 0.259***<br>(0.036) | −0.047***<br>(0.010) |
| <i>Panel D: 2SLS with Controls</i>    |                            |                           |                     |                         |                     |                      |
| Fitted Vac Rate                       | 0.855***<br>(0.099)        | 0.504***<br>(0.071)       | 0.057<br>(0.284)    | 0.002<br>(0.122)        | 0.257***<br>(0.036) | −0.048***<br>(0.010) |
| Num.Obs.                              | 899 285                    | 798 924                   | 342 056             | 527 970                 | 1 405 387           | 876 825              |
| FE: date                              | X                          | X                         | X                   | X                       | X                   | X                    |
| FE: county                            | X                          | X                         | X                   | X                       | X                   | X                    |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01

Table 2: Google Mobility—Effects of Vaccination on Mobility

Note: This table presents the estimated effects of COVID vaccine coverage on six mobility indicators. The outcome variables are from the Google Mobility data. Columns (1) to (5) present the effects of COVID vaccination on visits in public areas including: (1) retail & recreation areas, (2) grocery & pharmacies, (3) parks, (4) transit stations, and (5) workplaces. Column (6) presents the effects of COVID vaccination on the duration of stay in residential areas. Panels A and B present the OLS estimates. Panels C and D present the 2SLS estimates. Covariates including race, education, and poverty rates are included in Panels B and D. County and day fixed effects are included in all columns. Standard errors clustered at the county level are included in the parentheses. The estimation period is from December 13, 2020 to June 16, 2022.

|                               | (1)                                  | (2)                 | (3)                 | (4)              | (5)               |
|-------------------------------|--------------------------------------|---------------------|---------------------|------------------|-------------------|
|                               | Labor Force Participation Rate       |                     |                     |                  |                   |
|                               | reduced form                         | OLS                 |                     | IV               |                   |
| SSIV                          | 6.844<br>(4.896)                     |                     |                     |                  |                   |
| Vac Rate                      |                                      | 2.579**<br>(1.163)  | 2.480**<br>(1.110)  |                  |                   |
| Fitted Vac Rate               |                                      |                     |                     | 2.142<br>(1.466) | 1.745<br>(1.356)  |
| Num.Obs.                      | 918                                  | 918                 | 918                 | 918              | 918               |
| FE: year_month                | X                                    | X                   | X                   | X                | X                 |
| FE: state                     | X                                    | X                   | X                   | X                | X                 |
| Covariates:                   |                                      |                     | X                   |                  | X                 |
| IV Wald F-Stat:               |                                      |                     |                     | 14.9             | 15.3              |
|                               | Employment Rate                      |                     |                     |                  |                   |
| SSIV                          | 7.994<br>(5.607)                     |                     |                     |                  |                   |
| Vac Rate                      |                                      | 4.319***<br>(1.084) | 3.970***<br>(1.020) |                  |                   |
| (4762.981)<br>Fitted Vac Rate |                                      |                     |                     | 2.502<br>(1.753) | 2.188<br>(1.625)  |
| Num.Obs.                      | 918                                  | 918                 | 918                 | 918              | 918               |
| FE: year_month                | X                                    | X                   | X                   | X                | X                 |
| FE: state                     | X                                    | X                   | X                   | X                | X                 |
| Covariates:                   |                                      |                     | X                   |                  | X                 |
| IV Wald F-Stat:               |                                      |                     |                     | 14.9             | 15.3              |
|                               | Business Applications per 1,000      |                     |                     |                  |                   |
| SSIV                          | 0.135<br>(1.037)                     |                     |                     |                  |                   |
| Vac Rate                      |                                      | -0.416<br>(0.321)   | -0.435<br>(0.319)   |                  |                   |
| Fitted Vac Rate               |                                      |                     |                     | 0.042<br>(0.325) | 0.175<br>(0.328)  |
| Num.Obs.                      | 918                                  | 918                 | 918                 | 918              | 918               |
| FE: year_month                | X                                    | X                   | X                   | X                | X                 |
| FE: state                     | X                                    | X                   | X                   | X                | X                 |
| Covariates:                   |                                      |                     | X                   |                  | X                 |
| IV Wald F-Stat:               |                                      |                     |                     | 14.9             | 15.3              |
|                               | Home Mortgage Applications per 1,000 |                     |                     |                  |                   |
|                               | reduced form                         | OLS                 |                     | IV               |                   |
| SSIV                          | 0.152<br>(1.097)                     |                     |                     |                  |                   |
| Vac Rate                      |                                      | -0.019<br>(0.106)   | 0.051<br>(0.119)    |                  |                   |
| Fitted Vac Rate               |                                      |                     |                     | 0.047<br>(0.343) | -0.085<br>(0.371) |
| Num.Obs.                      | 918                                  | 918                 | 918                 | 918              | 918               |
| FE: year_month                | X                                    | X                   | X                   | X                | X                 |
| FE: state                     | X                                    | X                   | X                   | X                | X                 |
| Covariates:                   |                                      |                     | X                   |                  | X                 |
| IV Wald F-Stat:               |                                      |                     |                     | 14.9             | 15.3              |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01

**Table 3: Effects of COVID Vaccination on Economic Activities**

Note: This table presents the estimated effects of COVID vaccine coverage on four economic indicators, including the labor force participation rate, employment rate, the new business application rate per 1,000 population, and the home mortgage application rate per 1,000 population. The estimation closely follows the baseline analysis using state-by-month data. Since the time unit of observations is monthly, the explanatory variables are included contemporaneously rather than in lagged forms. Column (1) present the reduced form estimates. Columns (2) and (3) present the OLS estimates. Columns (4) and (5) present the IV estimates. Covariates including race, education, and poverty rates are included in Columns (1), (3), and (5). State and month fixed effects are included in all columns. Standard errors clustered at the state level are included in the parentheses. The estimation period is from December 2020 to May 2022. The estimates are multiplied by 100 for better legibility.

# 8 Figures

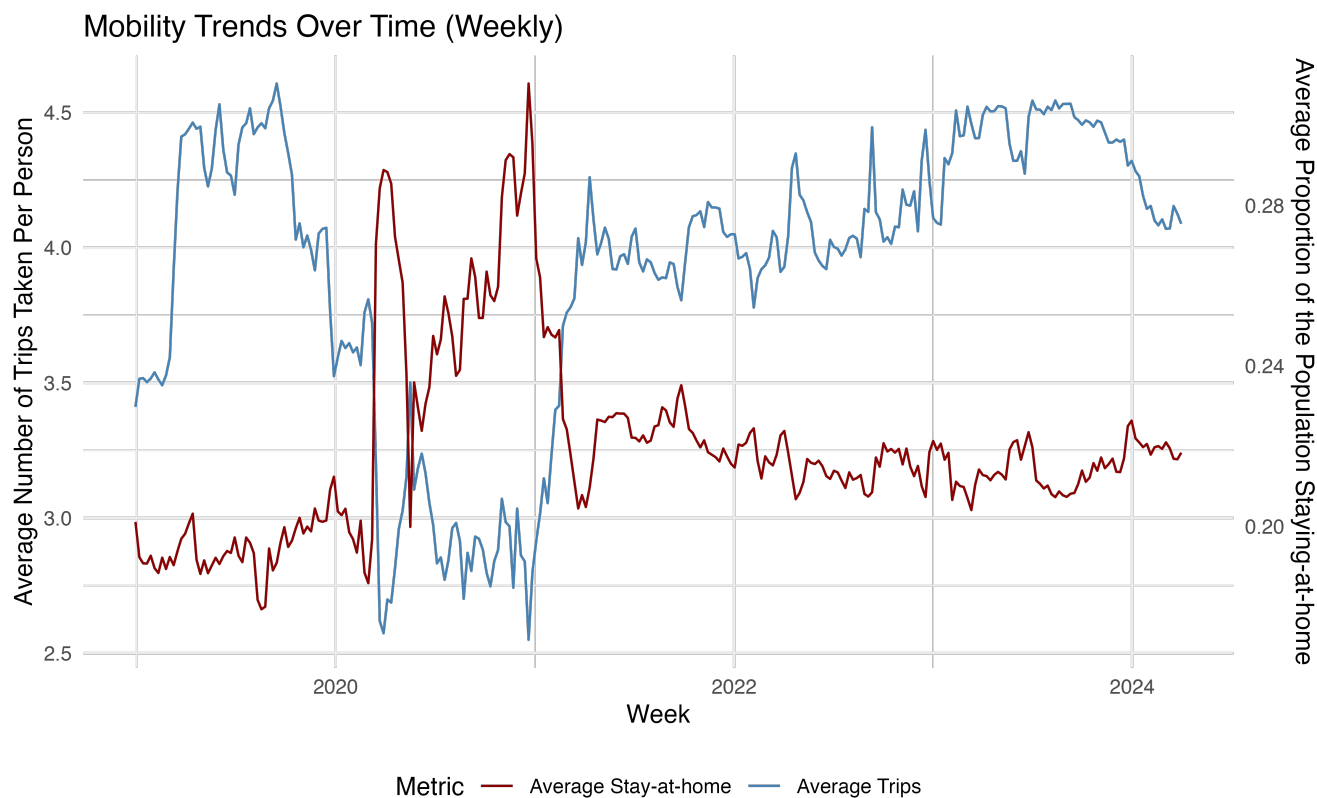
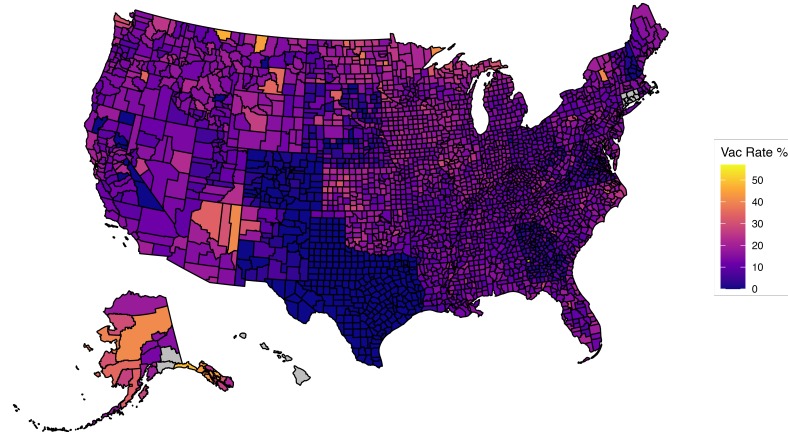


Figure 1: Mobility Trends

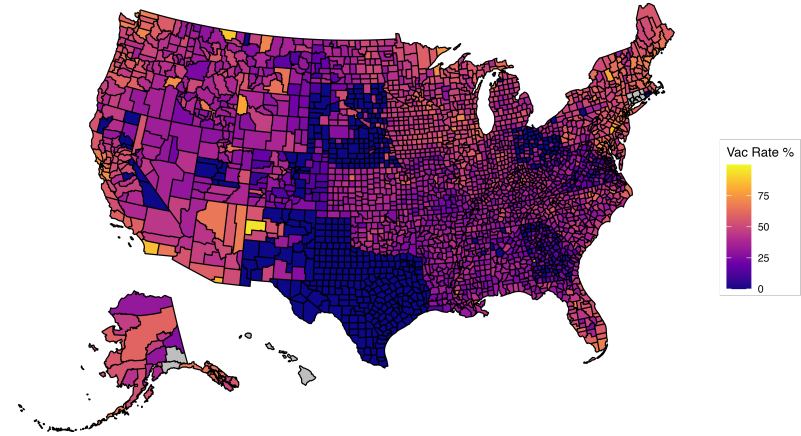
Note: This figure present the national trends for the weekly average number of trips taken per person and the weekly average proportion of the population staying-at-home. Source: BTS.

First Dose Vaccine Administration Rate by County 2021-02-28



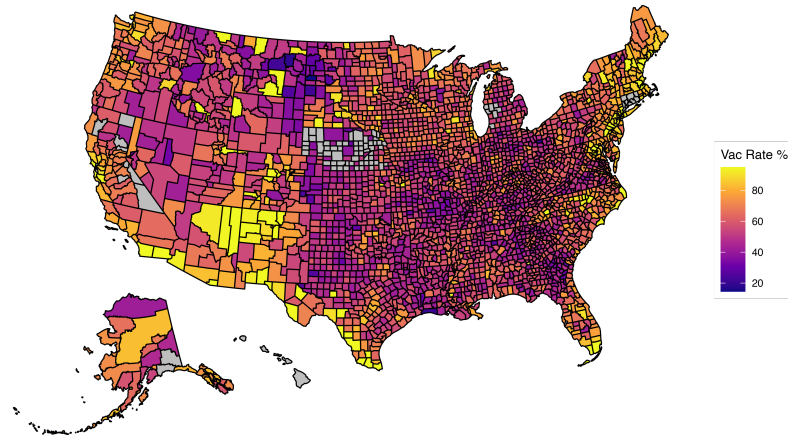
Feb 28, 2021

First Dose Vaccine Administration Rate by County 2021-07-03



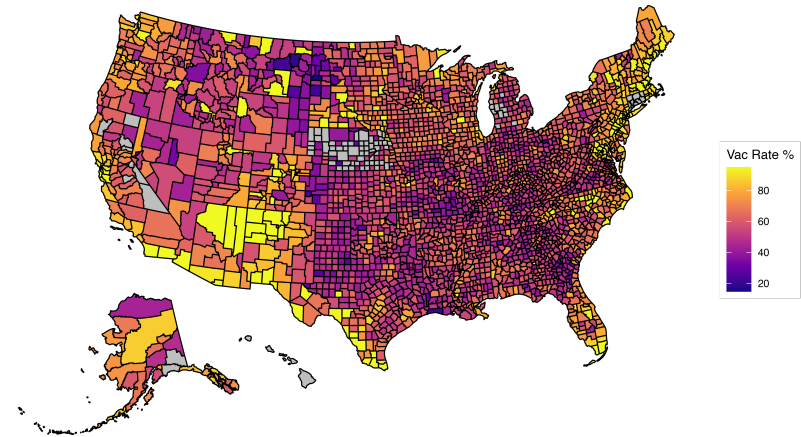
Jul 3, 2021

First Dose Vaccine Administration Rate by County 2022-06-16



Jun 16, 2022

First Dose Vaccine Administration Rate by County 2023-05-10



May 10, 2023

Figure 2: Snapshots for Vaccination Rates by County Over Time

Note: These figures plot the first-dose COVID vaccine administration rates on February 28, 2021; July 3, 2021; June 16, 2022; and May 10, 2023. The vaccination rates are capped at 100 percent for better visualization. Source: CDC.

## **A Appendix**

### **A.1 Summary Statistics**

|                                                           | Mean    | SD     | Min      | Median  | Max     |
|-----------------------------------------------------------|---------|--------|----------|---------|---------|
| Median age                                                | 41.537  | 5.386  | 21.400   | 41.300  | 68.900  |
| Proportion of the population identifying as White         | 0.750   | 0.202  | 0.006    | 0.822   | 1.000   |
| Proportion of the population without a highschool degree  | 0.082   | 0.040  | 0.000    | 0.074   | 0.540   |
| Poverty rate                                              | 0.136   | 0.057  | 0.000    | 0.128   | 0.568   |
| Retail and recreation visits percent change from baseline | -0.351  | 19.015 | -100.000 | -1.000  | 409.000 |
| Grocery and pharmacy visits percent change from baseline  | 4.590   | 16.556 | -100.000 | 4.000   | 332.000 |
| Parks visits percent change from baseline                 | 23.326  | 58.003 | -96.000  | 10.000  | 670.000 |
| Transit stations visits percent change from baseline      | -2.821  | 33.792 | -100.000 | -4.000  | 478.000 |
| Workplaces visits percent change from baseline            | -18.016 | 14.029 | -100.000 | -17.000 | 120.000 |
| Residential duration of stay percent change from baseline | 4.793   | 4.526  | -29.000  | 4.000   | 46.000  |
| Vaccination rate—female under 5                           | 0.075   | 0.026  | 0.002    | 0.084   | 0.101   |
| Vaccination rate—female 5 to 17                           | 0.291   | 0.206  | 0.000    | 0.285   | 0.562   |
| Vaccination rate—female 18 to 24                          | 0.575   | 0.280  | 0.000    | 0.685   | 0.871   |
| Vaccination rate—female 25 to 39                          | 0.596   | 0.262  | 0.000    | 0.702   | 0.860   |
| Vaccination rate—female 40 to 49                          | 0.660   | 0.273  | 0.000    | 0.779   | 0.902   |
| Vaccination rate—female 50 to 64                          | 0.721   | 0.285  | 0.000    | 0.839   | 0.959   |
| Vaccination rate—female 65 to 74                          | 0.851   | 0.289  | 0.000    | 0.951   | 1.079   |
| Vaccination rate—female over 75                           | 0.805   | 0.251  | 0.000    | 0.885   | 1.001   |
| Vaccination rate—male under 5                             | 0.075   | 0.026  | 0.002    | 0.084   | 0.100   |
| Vaccination rate—male 5 to 17                             | 0.276   | 0.198  | 0.000    | 0.268   | 0.537   |
| Vaccination rate—male 18 to 24                            | 0.500   | 0.258  | 0.000    | 0.610   | 0.766   |
| Vaccination rate—male 25 to 39                            | 0.541   | 0.255  | 0.000    | 0.647   | 0.793   |
| Vaccination rate—male 40 to 49                            | 0.611   | 0.273  | 0.000    | 0.730   | 0.857   |
| Vaccination rate—male 50 to 64                            | 0.689   | 0.291  | 0.000    | 0.811   | 0.933   |
| Vaccination rate—male 65 to 74                            | 0.862   | 0.296  | 0.000    | 0.963   | 1.101   |
| Vaccination rate—male over 75                             | 0.859   | 0.266  | 0.000    | 0.935   | 1.079   |
| Average N. of trips per person                            | 3.917   | 0.866  | 0.000    | 3.863   | 62.937  |
| Proportion of the population staying-at-home (%)          | 18.441  | 4.554  | 2.662    | 18.022  | 91.489  |
| Average N. of trips under 1 mile per person               | 0.929   | 0.382  | 0.000    | 0.888   | 23.079  |
| Average N. of trips 1 to 3 miles per person               | 0.898   | 0.300  | 0.000    | 0.882   | 9.722   |
| Average N. of trips 3 to 5 miles per person               | 0.433   | 0.216  | 0.000    | 0.427   | 21.225  |
| Average N. of trips 5 to 10 miles per person              | 0.575   | 0.252  | 0.000    | 0.559   | 11.685  |
| Average N. of trips 10 to 25 miles per person             | 0.629   | 0.283  | 0.000    | 0.602   | 58.882  |
| Average N. of trips 25 to 50 miles per person             | 0.289   | 0.183  | 0.000    | 0.261   | 13.694  |
| Average N. of trips 5 to 100 miles per person             | 0.114   | 0.087  | 0.000    | 0.096   | 6.349   |
| Average N. of trips 100 to 250 miles per person           | 0.042   | 0.031  | 0.000    | 0.035   | 1.297   |
| Average N. of trips 250 to 500 mile per person            | 0.006   | 0.007  | 0.000    | 0.005   | 0.880   |
| Average N. of trips over 500 miles per person             | 0.002   | 0.006  | 0.000    | 0.002   | 1.567   |
| Shift-share instrument with 2019 shares                   | 56.480  | 24.987 | 0.000    | 65.210  | 95.943  |
| Vaccination rate (%)                                      | 40.676  | 24.847 | 0.000    | 44.464  | 482.897 |
| Adjusted vaccination rate (%)                             | 41.201  | 24.707 | 0.000    | 44.686  | 477.903 |
| Labor market participation rate (%)                       | 62.604  | 3.838  | 53.500   | 62.500  | 71.900  |
| Employment rate (%)                                       | 60.097  | 3.996  | 50.100   | 59.900  | 68.800  |
| Business application per 1,000                            | 1.350   | 0.849  | 0.428    | 1.110   | 8.4339  |
| Home mortgage application per 1,000                       | 1.664   | 0.579  | 0.056    | 1.690   | 3.625   |

Table A1: Summary Statistics

## A.2 Addressing Sources of Measurement Errors

According to [Archived Methodology of COVID Vaccine Data](#), the first dose COVID vaccine coverage might be inflated due to the lack of linkage between follow-up doses and initial doses. As a result, the follow-up doses might be erroneously counted as the first doses. Figure A1 plots the national first-dose vaccine coverage rate by age and sex, showing that some groups had vaccination rate over 100%. There are two implications on the overcounting of the first-dose vaccination rate. First, if the overcounting is similar across all groups, it simply suggests that my regression estimates are conservative, as the magnitudes of the explanatory variable are overstated. Second, if the overcounting is not random (eg. there exists systematic biases), my coefficient estimates would be biased. For example, if a county was recorded to have a higher first-dose administration rate but lower second-dose administration rate compared to the national average, it is likely that the first-dose vaccination rate has been overcounted. To address the later issue, I assume that the chances of getting follow-up doses after getting the first dose is uniform across the nation, while varying across time.<sup>12</sup> This strategy relies on the assumption that the total number of vaccines distributed are better recorded compared to the first-dose vaccines. I calculate the predicted first dose vaccination rate,

$$\widetilde{VAC}_{it}^{l=1} = \underbrace{\sum_{l=1}^3 Vac_{it}}_{\text{Total doses of vaccine administered}} \times \underbrace{\frac{VAC_t^{l=1}}{\sum_l VAC_t}}_{\text{National share of the first doses}} \quad (5)$$

where  $VAC$  denotes vaccination rate of dose sequence,  $l$ . Thus, I predict the first dose vaccination rate  $\widetilde{VAC}_{it}^{l=1}$  in county  $i$  and time  $t$  using a uniform “propensity” of the first doses out of total doses. This design preserves the variation in vaccine uptake across counties while adjusting for the overcounting of first doses. The estimation then follows,

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<sup>12</sup>Especially, I keep the dose sequence up to three doses. I assume a complete vaccination series includes two doses. While Johnson&Johnson vaccine only requires one dose for a completed series, it is safe to assume this since COVID vaccines by Johnson&Johnson only made up 3% of the total doses distributed as of May 10, 2023 (see [COVID-19 Vaccinations in the United States, Jurisdiction by CDC](#)).

$$y_{it} = \beta \times \widehat{VAC}_{i,t-1}^{l=1} + \mathbf{X}'_{it} + \gamma_i + \delta_t + \epsilon_{it} \quad (6)$$

where  $\widehat{VAC}_{it}^{k=1}$  is obtained from the first-stage regression,

$$\widetilde{VAC}_{i,t-1}^{l=1} = \alpha_1 \widetilde{Z}_{i,t-1} + \mathbf{X}'_{i,t-1} + \theta_i + \iota_{t-1} + v_{i,t-1} \quad (7)$$

where the notations strictly follow the baseline Equations 3 and 4. This design plausibly reduces the worry on the systematic overcounting of the first dose vaccines and provides a conservative estimate.

The estimation results are presented in Table A2. I observe similar results to my baseline analysis, where COVID vaccine coverage led to an increase in the average number of trips taken per person and a decrease in the proportion of the population staying-at-home. These results bring external support to my primary finding that COVID vaccine coverage increased mobility.



|                          | (1)                                   | (2)                  | (3)                  | (4)                  | (5)                  |
|--------------------------|---------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                          | Trips per capita                      |                      |                      |                      |                      |
|                          | reduced form                          | OLS                  |                      | IV                   |                      |
| SSIV                     | 0.042***<br>(0.004)                   |                      |                      |                      |                      |
| Vac Rate Adjusted        |                                       | 0.005***<br>(0.001)  | 0.005***<br>(0.001)  |                      |                      |
| Fitted Vac Rate Adjusted |                                       |                      |                      | 0.039***<br>(0.005)  | 0.039***<br>(0.005)  |
| Num.Obs.                 | 1 719 712                             | 1 659 659            | 1 659 659            | 1 659 659            | 1 659 659            |
| FE: date                 | X                                     | X                    | X                    | X                    | X                    |
| FE: county               | X                                     | X                    | X                    | X                    | X                    |
| Covariates:              |                                       |                      | X                    |                      | X                    |
| IV Wald F-Stat:          |                                       |                      |                      | 96.4                 | 95.9                 |
|                          | (1)                                   | (2)                  | (3)                  | (4)                  | (5)                  |
|                          | Percentage of Population Stay-at-home |                      |                      |                      |                      |
|                          | reduced form                          | OLS                  |                      | IV                   |                      |
| SSIV                     | -0.155***<br>(0.021)                  |                      |                      |                      |                      |
| Vac Rate                 |                                       | -0.030***<br>(0.003) | -0.030***<br>(0.003) |                      |                      |
| Fitted Vac Rate Adjusted |                                       |                      |                      | -0.142***<br>(0.022) | -0.143***<br>(0.022) |
| Num.Obs.                 | 1 719 712                             | 1 659 659            | 1 659 659            | 1 659 659            | 1 659 659            |
| FE: date                 | X                                     | X                    | X                    | X                    | X                    |
| FE: county               | X                                     | X                    | X                    | X                    | X                    |
| Covariates:              |                                       |                      | X                    |                      | X                    |
| IV Wald F-Stat:          |                                       |                      |                      | 96.4                 | 95.9                 |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01

**Table A2: Effects of Vaccination on Trips and the Proportion of Population Staying-at-home**

Note: This table presents the effects of vaccination on the average number of trips taken per person and the percentage of the population staying-at-home, accounting for adjustments to first-dose coverage. Column (1) present the reduced form estimates. Columns (2) and (3) present the OLS estimates. Columns (4) and (5) present the IV estimates. Covariates including race, education, and poverty rates are included in Columns (1), (3), and (5). County and day fixed effects are included in all columns. Standard errors clustered at the county level are included in the parentheses. The estimation period is from December 13, 2020 to June 16, 2022.

### A.3 Effects by Trip Distance

It is important to examine the effects of COVID vaccination on the average number of trips taken per person by trip distance. Because, as previously argued, my estimates could be confounded with trips for vaccination. If this is the case, the economic implication of my estimates are compromised. Secondly, it is insightful to assess the heterogeneous impact of vaccine coverage on trips by distance. Exploiting the fact that trips for vaccination are generally short, I address this concern by adopting additional outcomes that measure the number of trips by different distance (e.g. It is unlikely to take over 100 miles for vaccination).

The results are presented in Table A3. I find consistent evidence using both OLS and 2SLS models that COVID vaccination led to an increase in trips, including trips of up to 250 miles.

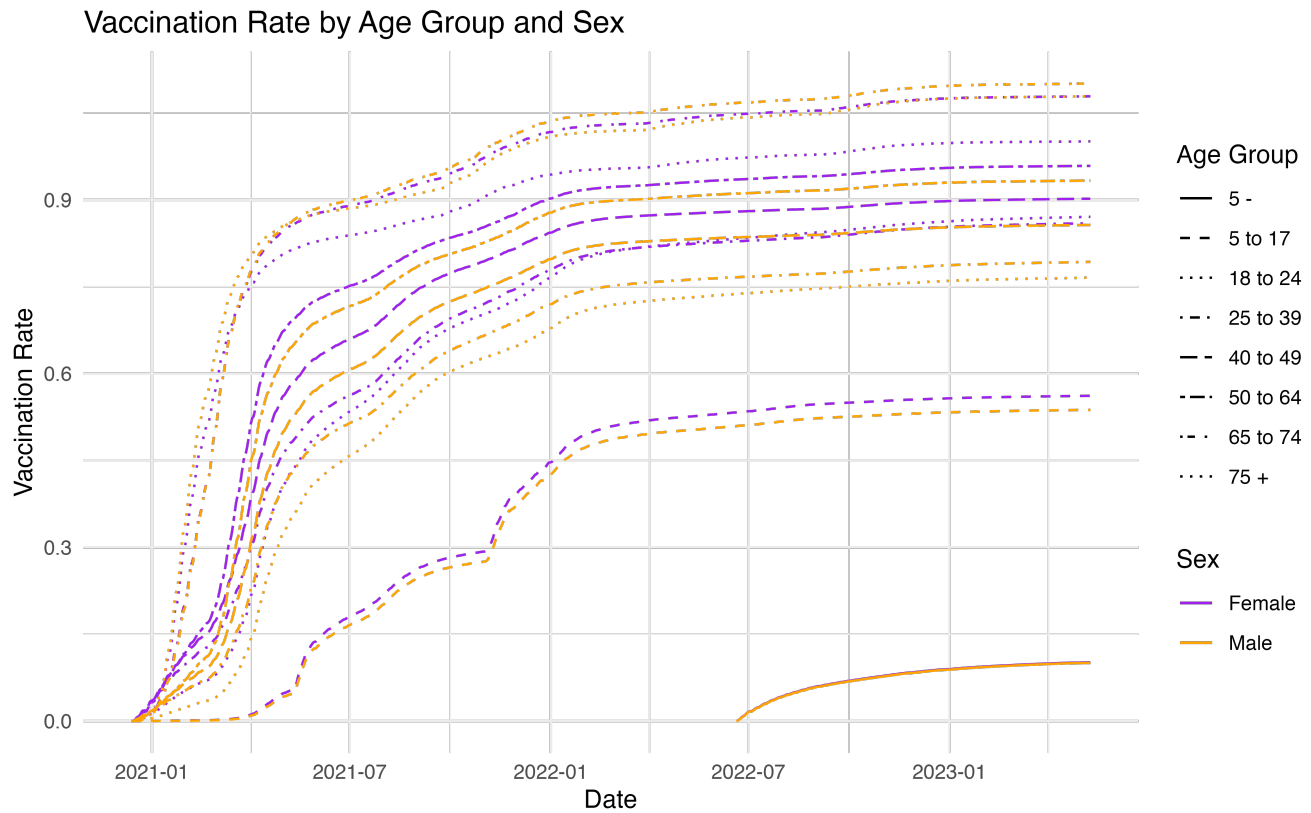


Figure A1: First-dose Vaccine Coverage by Age and Sex

While I find an increase in trips of over 250 miles using OLS models, the evidence is limited from 2SLS results. In summary, I find that COVID vaccine coverage increases the number of trips across all distance categories up to 250 miles. These results further provide confidence to my baseline findings.

| Trip Types                            | < 1 mile<br>(1)     | 1-3 miles<br>(2)    | 3-5 miles<br>(3)    | 5-10 miles<br>(4)   | 10-25 miles<br>(5)  | 25-50 miles<br>(6)  | 50-100 miles<br>(7) | 100-250 miles<br>(8) | 250-500 miles<br>(9) | > 500 miles<br>(10) |
|---------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|----------------------|----------------------|---------------------|
| <b>Panel A: OLS without Controls</b>  |                     |                     |                     |                     |                     |                     |                     |                      |                      |                     |
| Vac Rate                              | 0.187***<br>(0.026) | 0.058***<br>(0.014) | 0.018**<br>(0.007)  | 0.043***<br>(0.009) | 0.075***<br>(0.011) | 0.029***<br>(0.006) | 0.021***<br>(0.007) | 0.004***<br>(0.001)  | 0.001***<br>(0.000)  | 0.001***<br>(0.000) |
| <b>Panel B: OLS with Controls</b>     |                     |                     |                     |                     |                     |                     |                     |                      |                      |                     |
| Vac Rate                              | 0.187***<br>(0.026) | 0.058***<br>(0.014) | 0.018**<br>(0.008)  | 0.043***<br>(0.009) | 0.075***<br>(0.011) | 0.029***<br>(0.006) | 0.021***<br>(0.007) | 0.004***<br>(0.001)  | 0.001***<br>(0.000)  | 0.001***<br>(0.000) |
| <b>Panel C: 2SLS without Controls</b> |                     |                     |                     |                     |                     |                     |                     |                      |                      |                     |
| Fitted Vac Rate                       | 1.224***<br>(0.198) | 0.905***<br>(0.140) | 0.448***<br>(0.083) | 0.512***<br>(0.084) | 0.763***<br>(0.123) | 0.200*<br>(0.102)   | 0.112**<br>(0.046)  | 0.092***<br>(0.014)  | 0.002<br>(0.002)     | −0.009<br>(0.006)   |
| <b>Panel D: 2SLS with Controls</b>    |                     |                     |                     |                     |                     |                     |                     |                      |                      |                     |
| Fitted Vac Rate                       | 1.223***<br>(0.198) | 0.900***<br>(0.139) | 0.452***<br>(0.084) | 0.510***<br>(0.083) | 0.759***<br>(0.122) | 0.200*<br>(0.104)   | 0.111**<br>(0.046)  | 0.092***<br>(0.014)  | 0.002<br>(0.002)     | −0.008<br>(0.006)   |
| Num.Obs.                              | 1 665 914           | 1 665 914           | 1 665 914           | 1 665 914           | 1 665 914           | 1 665 914           | 1 665 914           | 1 665 914            | 1 665 914            | 1 665 914           |
| FE: date                              | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                    | X                    | X                   |
| FE: county                            | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                    | X                    | X                   |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01

**Table A3: Effects of Vaccination on Trips by Trip Distance**

Note: This table presents the effects of vaccination coverage on the daily number of trips per person by trip distance. The data for the outcome variables comes from BTS mobility statistics. Panels A and B present the OLS estimates without and with controls, respectively. Panels C and D present the 2SLS estimates without and with controls, respectively. Covariates including race, education, and poverty rates are included in Panels B and D. County and day fixed effects are included for all estimates. Standard errors clustered at the county level are included in the parentheses. The estimation period is from December 13, 2020 to June 16, 2022. The estimates are multiplied by 100 for better legibility.

## A.4 Effects by COVID Wave

It is important to examine the effects of vaccination coverage on mobility across different waves of COVID-19 for two main reasons. First, one can examine the effects of COVID vaccination while holding the severity of infection comparable. Second, it is important to assess the heterogeneous effects of vaccination coverage in different phases of the pandemic. To visualize COVID waves, I plot the trends of weekly deaths in Figure A2. In which, the red dotted line denotes the beginning and the end of data collecting periods from December 13, 2020 to May 10, 2023. Additionally, I break the data on July 5, 2021, where the new COVID deaths reached a lowest point since the start of the pandemic, and November 13, 2021, where new COVID deaths reached a visible trough. Since June 16, 2022, CDC began to update data weekly. To maintain consistency, I exclude the periods after June 16, 2022 from the analysis sample.

I present the results in Table A4. Columns (1) and (4) present the estimated effects of vaccination coverage on outcomes—the number of trips taken per person and the percentage of population staying at home, respectively—during the period from December 13, 2020, to July 3, 2021, corresponding to the periods from the initial roll-out of COVID vaccine to the first trough of new deaths since vaccination began. Binned by COVID wave, Columns (2 and (5) present the estimated effects from July 4, 2021, to November 13, 2021, and Columns (3) and (6) present the estimated effects from November 14, 2021 to June 16, 2022. Panels A through D present the results from OLS without controls, OLS with controls, 2SLS without controls and 2SLS with controls, respectively.

According to Columns (1) and (4), I find statistically significant results with “expected” signs (positive effects of trips and negative effects on the percentage of individuals staying-at-home), consistent with my baseline analysis. Turning into Columns (2) and (5), I find that, OLS estimates exhibit the opposite signs to the 2SLS estimates. While the OLS estimates suggest that COVID vaccine coverage reduced mobility during the periods from July 4, 2021, to November 13, 2021, the 2SLS estimates suggest that COVID vaccine led to an increase in mobility. Columns (4) and (6) presents the effects of COVID vaccine coverage from November 14, 2021,

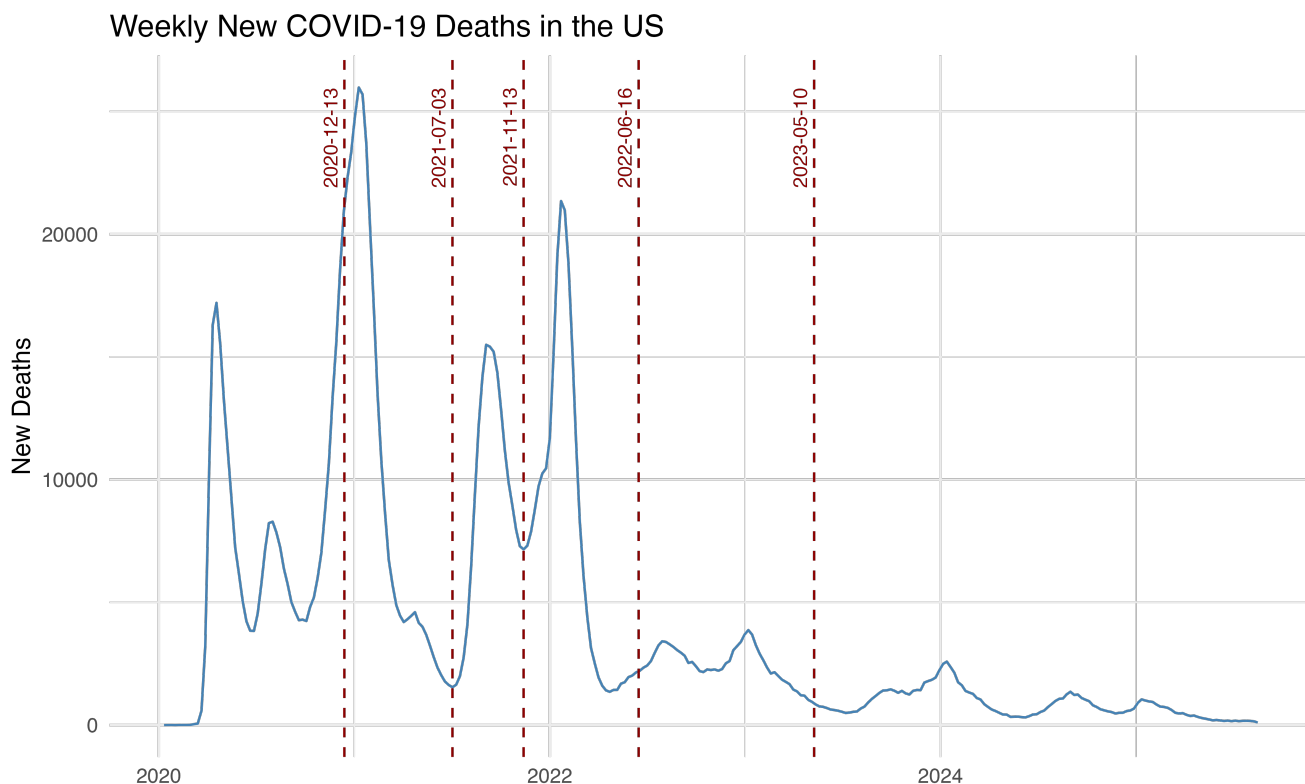


Figure A2: Weekly New COVID-19 Deaths

Note: This figure plots the weekly new COVID-19 deaths in the US. Source: CDC.

to June 16, 2022, and the results consistently suggest that COVID vaccine coverage increased mobility in these periods. However, the interpretation of the estimates in periods from July 4, 2021, to June 16, 2022, requires extra scrutiny, as the respected F-stat dropped to 22.2 and 15.0 while controls are included. While the values of the F-stat are still above the rule-of-thumb  $F > 10$ , recent research suggests that the critical value required may be substantially higher (Keane and Neal, 2023; Young, 2022).

In summary, the effects of COVID-19 vaccine coverage on mobility exhibit important heterogeneity. I find strong evidence that COVID vaccination coverage encouraged trips and reduced the likelihood of staying at home during the first wave since the vaccination began. While COVID vaccine coverage was still found to show positive effects on mobility after July 4, 2021, the first-stage F-stat declined noticeably during these periods, which deserves extra caution in interpreting the estimates.

|                                       | (1)                         | (2)                                             | (3)                         | (4)                                                                     | (5)                         | (6)                         |
|---------------------------------------|-----------------------------|-------------------------------------------------|-----------------------------|-------------------------------------------------------------------------|-----------------------------|-----------------------------|
|                                       | 2020-12-13<br>to 2021-07-03 | Trips per capita<br>2021-07-04<br>to 2021-11-13 | 2021-11-14<br>to 2022-06-16 | Percentage of Population Staying-at-home<br>2020-12-13<br>to 2021-07-03 | 2021-07-04<br>to 2021-11-13 | 2021-11-14<br>to 2022-06-16 |
| <b>Panel A: OLS without Controls</b>  |                             |                                                 |                             |                                                                         |                             |                             |
| Vac Rate                              | 0.007***<br>(0.000)         | −0.003***<br>(0.000)                            | 0.002***<br>(0.001)         | −0.057***<br>(0.004)                                                    | 0.006***<br>(0.001)         | −0.004*<br>(0.002)          |
| <b>Panel B: OLS with Controls</b>     |                             |                                                 |                             |                                                                         |                             |                             |
| Vac Rate                              | 0.007***<br>(0.000)         | −0.003***<br>(0.000)                            | 0.002***<br>(0.001)         | −0.057***<br>(0.004)                                                    | 0.006***<br>(0.001)         | −0.004**<br>(0.002)         |
| <b>Panel C: 2SLS without Controls</b> |                             |                                                 |                             |                                                                         |                             |                             |
| Fitted Vac Rate                       | 0.014***<br>(0.003)         | 0.061***<br>(0.016)                             | 0.073***<br>(0.020)         | −0.059***<br>(0.020)                                                    | −0.130***<br>(0.035)        | −0.467***<br>(0.125)        |
| IV Wald F-Stat:                       | 60.9                        | 22.2                                            | 14.3                        | 60.9                                                                    | 22.2                        | 14.3                        |
| <b>Panel D: 2SLS with Controls</b>    |                             |                                                 |                             |                                                                         |                             |                             |
| Fitted Vac Rate                       | 0.014***<br>(0.003)         | 0.061***<br>(0.016)                             | 0.077***<br>(0.021)         | −0.059***<br>(0.020)                                                    | −0.130***<br>(0.035)        | −0.477***<br>(0.124)        |
| IV Wald F-Stat:                       | 61.1                        | 22.2                                            | 15.0                        | 61.1                                                                    | 22.2                        | 15.0                        |
| Num.Obs.                              | 615 007                     | 394 693                                         | 656 214                     | 615 007                                                                 | 394 693                     | 656 214                     |
| FE: date                              | X                           | X                                               | X                           | X                                                                       | X                           | X                           |
| FE: county                            | X                           | X                                               | X                           | X                                                                       | X                           | X                           |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01

**Table A4: Effects of Vaccination on Mobility by COVID Wave**

Note: This table presents the effects of COVID vaccine coverage on the average number of trips taken per person and the percentage of the population staying-at-home during different COVID waves. The data for the outcome variables comes from BTS mobility statistics. Columns (1) and (4) present the estimates using sample period from December 13, 2020, to July 3, 2021. Columns (2) and (5) present the estimates using sample period from July 4, 2021, to November 13, 2021. Columns (3) and (6) present the estimates using sample period from November 14, 2021, to June 16, 2022. Columns (1), (2), and (3) present the effects on the average number of trips taken per person. Columns (4), (5), and (6) present the effects on the percentage of the population staying-at-home. Panels A and B present the OLS estimates without and with controls, respectively. Panels C and D present the 2SLS estimates without and with controls, respectively. Covariates including race, education, and poverty rates are included in Panels B and D. County and day fixed effects are included for all estimates. Because the covariates are measured at the yearly level, they are omitted in Columns (2) and (5) as a result of collinearity. Standard errors clustered at the county level are included in the parentheses. The estimation period is from December 13, 2020 to June 16, 2022. The estimates are multiplied by 100 for better legibility.

## A.5 Effects by Presidential Election Vote Share

A concern about my identification is that that different age and sex groups may systematically differ in their political beliefs, which may in turn both impact their willingness to get vaccinated and their mobility patterns (see [Albrecht \(2022\)](#); [Barro \(2022\)](#); [Cao et al. \(2022\)](#)). I try to alleviate such concerns using subsample analysis in counties with similar political leanings. This exercise likely reduce the concerns about biases introduced through the correlation between political beliefs and the shares of the SSIV. I obtain the 2016 presidential election vote shares by county compiled by [MIT Election Data and Science Lab \(2018\)](#). I divide counties into four quantiles based on their partisan composition, ranging from those with the highest Republican vote shares to those with the highest Democratic vote shares. I then estimate the effects of COVID vaccine coverage on mobility following my main analysis for each of the four

quantiles.

I present the results in Table A5. Columns (1)-(4) and Columns (5)-(8) present the effects of COVID vaccine coverage on the average number of trips taken per capita and the percentage of the population staying-at-home, respectively. Columns (1)-(4) and Columns (5)-(8) each present the effects in counties with the lowest Democratic vote shares to the highest Democratic vote shares. It is worth noting the first-stage F statistics went below the rule-of-thumb level ( $F = 10$ ) for Columns (4) and (8), corresponding to counties with the highest Democratic vote shares. Thus, I focus on interpreting the results from other columns. According to Panels A and B, I find mild evidence that COVID vaccination increased the number of trips taken, and stronger evidence that COVID vaccination decreased the likelihood of staying at home. However, the OLS estimates can be biased due to endogeneity. I then turn to the IV estimates presented in Panels C and D. I find consistent evidence that COVID vaccine coverage increased the number of trips taken per capita and decreased the likelihood of staying at home. The results are significant at the 95% confidence level in Columns (1)-(3) and Columns (5)-(7), providing additional support for my main results.

|                                       | (1)                                   | (2)                 | (3)                 | (4)               | (5)                                                           | (6)                  | (7)                  | (8)                 |
|---------------------------------------|---------------------------------------|---------------------|---------------------|-------------------|---------------------------------------------------------------|----------------------|----------------------|---------------------|
|                                       | Trips per capita<br>R $\rightarrow$ D |                     |                     |                   | Percentage of Population Staying-at-home<br>R $\rightarrow$ D |                      |                      |                     |
|                                       | Q1                                    | Q2                  | Q3                  | Q4                | Q1                                                            | Q2                   | Q3                   | Q4                  |
| <b>Panel A: OLS without Controls</b>  |                                       |                     |                     |                   |                                                               |                      |                      |                     |
| Vac Rate                              | 0.000<br>(0.001)                      | 0.000<br>(0.001)    | 0.002**<br>(0.001)  | 0.001*<br>(0.001) | -0.009*<br>(0.005)                                            | -0.017***<br>(0.004) | -0.022***<br>(0.004) | -0.009**<br>(0.004) |
| <b>Panel B: OLS with Controls</b>     |                                       |                     |                     |                   |                                                               |                      |                      |                     |
| Vac Rate                              | 0.000<br>(0.001)                      | 0.000<br>(0.001)    | 0.002***<br>(0.001) | 0.001*<br>(0.001) | -0.009*<br>(0.005)                                            | -0.016***<br>(0.004) | -0.022***<br>(0.004) | -0.010**<br>(0.004) |
| <b>Panel C: 2SLS without Controls</b> |                                       |                     |                     |                   |                                                               |                      |                      |                     |
| Fitted Vac Rate                       | 0.047***<br>(0.015)                   | 0.034***<br>(0.007) | 0.028***<br>(0.007) | 0.060*<br>(0.034) | -0.110**<br>(0.043)                                           | -0.223***<br>(0.044) | -0.177***<br>(0.033) | -0.348*<br>(0.211)  |
| IV Wald F-Stat:                       | 24.2                                  | 43.9                | 36.6                | 3.0               | 24.2                                                          | 43.9                 | 36.6                 | 3.0                 |
| <b>Panel D: 2SLS with Controls</b>    |                                       |                     |                     |                   |                                                               |                      |                      |                     |
| Fitted Vac Rate                       | 0.047***<br>(0.015)                   | 0.034***<br>(0.007) | 0.029***<br>(0.007) | 0.057*<br>(0.030) | -0.114**<br>(0.044)                                           | -0.222***<br>(0.043) | -0.179***<br>(0.034) | -0.325*<br>(0.187)  |
| IV Wald F-Stat:                       | 24.3                                  | 45.1                | 36.1                | 3.4               | 24.3                                                          | 45.1                 | 36.1                 | 3.4                 |
| Num.Obs.                              | 437 647                               | 454 395             | 457 862             | 445 898           | 437 647                                                       | 454 395              | 457 862              | 445 898             |
| FE: date                              | X                                     | X                   | X                   | X                 | X                                                             | X                    | X                    | X                   |
| FE: county                            | X                                     | X                   | X                   | X                 | X                                                             | X                    | X                    | X                   |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01

**Table A5: Effects of Vaccination on Mobility by Presidential Vote**

Note: This table presents the effects of COVID vaccine coverage on the average number of trips taken per person and the percentage of the population staying-at-home by the quantile of presidential vote. The data for the outcome variables comes from BTS mobility statistics. Columns (1)–(4) and Columns (5)–(8) report results by quartiles of Democratic vote share. That is, Columns (1)–(4) and Columns (5)–(8) correspond to counties ranging from those that are predominantly Republican to those that are predominantly Democratic. Columns (1)–(4) present the effects on the average number of trips taken per person. Columns (5)–(8) present the effects on the percentage of the population staying-at-home. Panels A and B present the OLS estimates without and with controls, respectively. Panels C and D present the 2SLS estimates without and with controls, respectively. Covariates including race, education, and poverty rates are included in Panels B and D. County and day fixed effects are included for all estimates. Standard errors clustered at the county level are included in the parentheses. The estimation period is from December 13, 2020 to June 16, 2022.



## A.6 Using Mobility Changes as Outcomes

A further exercise I adopt is that I examine the effects of vaccine coverage on the *changes* of mobility, defined as the first difference of mobility. I estimate using the same equations for instrument construction and first stage in Equations 2 and 4. The second stage could be stated as follows,

$$\delta y_{it} = \beta \times \widehat{VAC}_{i,t-1} + \mathbf{X}'_{it} + \gamma_i + \delta_t + \epsilon_{it} \quad (8)$$

where  $\delta y_{it}$  denotes the changes in mobility.<sup>13</sup> With this exercise, I exploit the arguably orthogonal relationship between age and sex composition and mobility trends to examine the validity of the instruments. If this exercise yields estimated effects of “expected” signs, this would give me more confidence to the main findings of this study.

I present the results in Table A6, where Column (1) presents the reduced form estimates, Columns (2) and (3) present OLS estimates, and Columns (4) and (5) present IV estimates. I find positive and significant effects of COVID vaccine coverage on the average number of trips taken per person. While the effects of COVID vaccine coverage on the proportion of the population staying-at-home loses significance in OLS results, I find consistent signs suggesting a negative effect. In summary, these findings provide further support to my baseline findings.

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<sup>13</sup>This is equivalent to  $y_{i,t} - y_{i,t-1}$ .

|                                      | (1)                                   | (2)                 | (3)                 | (4)                  | (5)                  |
|--------------------------------------|---------------------------------------|---------------------|---------------------|----------------------|----------------------|
|                                      | Trips per capita                      |                     |                     |                      |                      |
|                                      | reduced form                          | OLS                 |                     | IV                   |                      |
| SSIV                                 | 0.042***<br>(0.004)                   |                     |                     |                      |                      |
| Vac Rate                             |                                       | 0.004***<br>(0.001) | 0.004***<br>(0.001) |                      |                      |
| Fitted Vac Rate                      |                                       |                     |                     | 0.043***<br>(0.006)  | 0.043***<br>(0.006)  |
| Num.Obs.                             | 1 716 578                             | 1 665 914           | 1 665 914           | 1 665 914            | 1 665 914            |
| FE: date                             | X                                     | X                   | X                   | X                    | X                    |
| FE: county                           | X                                     | X                   | X                   | X                    | X                    |
| Covariates:                          | X                                     |                     | X                   |                      | X                    |
| IV Wald F statistics:                |                                       |                     |                     | 76.5                 | 76.1                 |
| * p < 0.1, ** p < 0.05, *** p < 0.01 |                                       |                     |                     |                      |                      |
|                                      | (1)                                   | (2)                 | (3)                 | (4)                  | (5)                  |
|                                      | Percentage of Population Stay-at-home |                     |                     |                      |                      |
|                                      | reduced form                          | OLS                 |                     | IV                   |                      |
| SSIV                                 | -0.238***<br>(0.049)                  |                     |                     |                      |                      |
| Vac Rate                             |                                       | -0.003<br>(0.002)   | -0.002<br>(0.002)   |                      |                      |
| Fitted Vac Rate                      |                                       |                     |                     | -0.229***<br>(0.054) | -0.231***<br>(0.054) |
| Num.Obs.                             | 1 714 707                             | 1 664 179           | 1 664 179           | 1 664 179            | 1 664 179            |
| FE: date                             | X                                     | X                   | X                   | X                    | X                    |
| FE: county                           | X                                     | X                   | X                   | X                    | X                    |
| Covariates:                          | X                                     |                     | X                   |                      | X                    |
| IV Wald F statistics:                |                                       |                     |                     | 76.5                 | 76.1                 |
| * p < 0.1, ** p < 0.05, *** p < 0.01 |                                       |                     |                     |                      |                      |

Table A6: Effects of Vaccination on Changes in Trips and the Proportion of Population Staying-at-home

Note: This table presents the estimated effects of COVID vaccine coverage on *changes* in the average number of trips taken per person and *changes* in the percentage of the population staying-at-home. Column (1) present the reduced form estimates. Columns (2) and (3) present the OLS estimates. Columns (4) and (5) present the IV estimates. Covariates including race, education, and poverty rates are included in Columns (1), (3), and (5). County and day fixed effects are included in all columns. Standard errors clustered at the county level are included in the parentheses. The estimation period is from December 13, 2020 to June 16, 2022. The estimates are multiplied by 100 for better legibility.

## A.7 Pre-trends

Following Goldsmith-Pinkham et al. (2020), I test whether the pre-determined demographics by age and sex were correlated with the outcome trends in pre-treatment periods. Shall the pre-determined demographic characteristics are exogenous, it is expected that they are un-correlated with outcomes, as the effects of treatment could only be materialized through shocks, which are absent by construction. Specifically, I regress trends of the pre-treatment outcome on the share for every age and sex group  $k$ ,

$$\Delta y_{it} = \beta^k \times \left\{ \frac{demgroup_i^{k,2019}}{pop_i^{k,2019}} \right\} + \mathbf{X}_i' + \lambda_t + \epsilon_{it}^k \quad (9)$$

where I obtain coefficients for each group. These include coefficients for the population shares of 0–4, 5–17, 18–24, 25–39, 40–49, 50–64, 65–74, and 75+ years, by male and female, yielding 16 groups in total.  $\Delta y_{it}$  captures changes in mobility, measured either as the difference between January 1 and December 31, 2019 (long difference), or as daily differences within 2019 (short difference).  $\frac{demgroup_i^{k,2019}}{pop_i^{k,2019}}$  denotes the demographic shares within county  $i$  in 2019,  $\mathbf{X}_i'$  denotes a vector of controls,  $\lambda_t$  denotes a time fixed effects (only when short difference is applied), and  $\epsilon_{it}^k$  denotes the error term.

I include the results in Table A7, where Columns (1) and (2) present the results on changes in daily trips using long- and short-difference forms. Columns (3) and (4) present the results on changes in the percentage of individuals staying-at-home using long- and short-difference forms. I find that the age and sex shares had little impacts on trips according to Columns (1) and (2), as most of the coefficients are statistically insignificant. Turning to Column (3), I find that the pre-determined shares predict the trends on the staying-at-home behavior. However, in Column (4) I find little evidence of such pattern. As a result, the pre-trend analysis shows little evidence that my instrument correlates with the trends of pre-treatment outcomes, which increases confidence on my shared-based identification method.

|                     | (1)                  | (2)                  | (3)                  | (4)                  |
|---------------------|----------------------|----------------------|----------------------|----------------------|
|                     | $\Delta Trips_i$     | $\delta Trips_{it}$  | $\Delta SAH_i$       | $\delta SAH_{it}$    |
| Male 5 and under    | 3.927<br>(2.967)     | 0.026<br>(0.022)     | -0.221***<br>(0.083) | -0.001<br>(0.001)    |
| Male 5-17           | 2.355*<br>(1.25)     | 0.015<br>(0.011)     | -0.117***<br>(0.041) | -0.001***<br>(0.000) |
| Male 18-24          | 0.064<br>(0.794)     | -0.001<br>(0.005)    | -0.008<br>(0.034)    | 0.000<br>(0.000)     |
| Male 25-39          | 2.486***<br>(0.749)  | 0.013***<br>(0.005)  | -0.105***<br>(0.026) | 0.000<br>(0.000)     |
| Male 40-49          | -1.067<br>(1.995)    | -0.012<br>(0.014)    | -0.151***<br>(0.053) | 0.000<br>(0.000)     |
| Male 50-64          | -1.283<br>(1.452)    | 0.003<br>(0.013)     | 0.224***<br>(0.044)  | 0.001<br>(0.000)     |
| Male 65-74          | 0.644<br>(1.364)     | -0.022*<br>(0.012)   | 0.114***<br>(0.039)  | 0.000<br>(0.000)     |
| Male 75 and above   | -3.344*<br>(1.753)   | 0.000<br>(0.016)     | 0.281***<br>(0.05)   | 0.001<br>(0.000)     |
| Female 5 and under  | 8.246**<br>(3.226)   | 0.059**<br>(0.025)   | -0.120<br>(0.075)    | -0.001*<br>(0.001)   |
| Female 5-17         | 1.168<br>(1.422)     | 0.016<br>(0.011)     | -0.141***<br>(0.042) | -0.001***<br>(0.000) |
| Female 18-24        | 0.049<br>(0.801)     | 0.003<br>(0.006)     | -0.029<br>(0.029)    | 0.000<br>(0.000)     |
| Female 25-39        | 1.779<br>(1.108)     | 0.004<br>(0.008)     | -0.245***<br>(0.035) | 0.000<br>(0.000)     |
| Female 40-49        | -1.578<br>(1.775)    | -0.033**<br>(0.016)  | -0.272***<br>(0.057) | 0.000<br>(0.000)     |
| Female 50-64        | -2.213<br>(1.404)    | -0.015<br>(0.011)    | 0.155***<br>(0.04)   | 0.001***<br>(0.000)  |
| Female 65-74        | -1.523<br>(1.438)    | -0.025***<br>(0.009) | 0.087**<br>(0.038)   | 0.000<br>(0.000)     |
| Female 75 and above | -8.275***<br>(1.154) | -0.017*<br>(0.009)   | 0.469***<br>(0.043)  | 0.001**<br>(0.000)   |

Table A7: Pre-trend Tests

Note: This table presents the pre-trend test results. The data for the outcome variables comes from BTS mobility statistics. The coefficients are estimated separately using the share of each age and sex group. Columns (1) and (2) present the reduced-form estimates on long- and short-differences measure of trips. Columns (3) and (4) present the reduced-form estimates on long- and short-differences measure of stay-at-home behaviors. Covariates including race, education, and poverty rates are included in all columns. day fixed effects are included in Columns (2) and (4). Heteroscedasticity robust standard errors are included in the parentheses for Columns (1) and (3). Standard errors clustered at the county level are included in the parentheses for Columns (2) and (4). The estimation period is from December 13, 2020 to June 16, 2022.

## A.8 Using Lagged Shares

A further strategy to check the robustness of my results are using lagged demographic makeup as the shares of the SSIV. By using lagged shares, it allows me to construct an instrument in spirit of the “past settlement” instrument that is more plausible to be exogenous (see (Jaeger et al., 2018; Borusyak et al., 2025)). This exercise gives more confidence to my results if the results are robust to lagged shares. Additionally, it allows me to evaluate the extent to which my estimates are sensitive to the the year-to-year changes of demographic composition. The shift-share IV could be written as,

$$\widetilde{Z}_{it} = \sum_k \underbrace{\frac{demgroup_{ik}^o}{pop_i^o}}_{shares} \times \underbrace{VAC_{kt}}_{shifts} \quad (10)$$

where  $o \in \{2011 : 2019\}$ , and other notations follow Equations (2).

I present the results using historical shares of demographic compositions as the shift-share IV components in Table A8. Columns (1) to (9) explore the effects of COVID vaccine coverage on the number of trips per person using shift-share IV constructed using demographic compositions during years from 2011 to 2019, respectively. The data on demographic compositions comes from the ACS. The results show consistency that vaccine coverage lead to an increase in trips, where all estimates are significant with the “expected” sign. However, one thing worth noticing is that the F-Stat of the first-stage regressions declined as the year of shares moved to a more distant past (the F-stat was at 32.6, higher than the  $F > 10$  level in 2015, while decreased to 4.7 moving to 2014), suggesting a diminished predictive power of historical shares on current demographic structure of vaccination. In summary, by alternating shares into a distant past, I find consistent results supporting the positive relationship between vaccine coverage and mobility.

| Year of Lagged Shares                   | 2011<br>(1)         | 2012<br>(2)         | 2013<br>(3)         | 2014<br>(4)         | 2015<br>(5)          | 2016<br>(6)          | 2017<br>(7)          | 2018<br>(8)          | 2019<br>(9)          |
|-----------------------------------------|---------------------|---------------------|---------------------|---------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| <b>Panel A: Trips</b>                   |                     |                     |                     |                     |                      |                      |                      |                      |                      |
| Fitted Vac Rate                         | 0.056***<br>(0.018) | 0.060***<br>(0.019) | 0.061***<br>(0.019) | 0.060***<br>(0.019) | 0.053***<br>(0.009)  | 0.049***<br>(0.008)  | 0.047***<br>(0.007)  | 0.046***<br>(0.006)  | 0.043***<br>(0.006)  |
| <b>Panel B: Stay-At-Home Population</b> |                     |                     |                     |                     |                      |                      |                      |                      |                      |
| Fitted Vac Rate                         | -0.271**<br>(0.112) | -0.262**<br>(0.111) | -0.246**<br>(0.106) | -0.225**<br>(0.103) | -0.236***<br>(0.045) | -0.213***<br>(0.037) | -0.195***<br>(0.032) | -0.177***<br>(0.028) | -0.164***<br>(0.026) |
| Num.Obs.                                | 1 665 844           | 1 665 844           | 1 665 844           | 1 665 844           | 1 665 844            | 1 665 844            | 1 665 844            | 1 665 844            | 1 665 844            |
| FE: date                                | X                   | X                   | X                   | X                   | X                    | X                    | X                    | X                    | X                    |
| FE: county                              | X                   | X                   | X                   | X                   | X                    | X                    | X                    | X                    | X                    |
| Covariates:                             | X                   | X                   | X                   | X                   | X                    | X                    | X                    | X                    | X                    |
| IV Wald F-Stat:                         | 4.5                 | 4.5                 | 4.6                 | 4.7                 | 32.6                 | 42.5                 | 53.4                 | 66.8                 | 74.9                 |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01

**Table A8: Effects of Vaccine Coverage on Mobility Using Alternative Shift-share IVs with Lagged Shares**

Note: This table presents the estimated effects of COVID vaccine coverage on the average number of trips taken per person and the percentage of the population staying-at-home using alternative shares. Columns (1)-(9) use county-level age and sex compositions from 2011 to 2019, respectively, as the shares of the SSIV. County and day fixed effects are included in all columns. Standard errors clustered at the county level are included in the parentheses. The estimation period is from December 13, 2020 to June 16, 2022. The estimates are multiplied by 100 for better legibility.

## A.9 Additional Tables & Figures

|                                                      | (1)                         | (2)                 |
|------------------------------------------------------|-----------------------------|---------------------|
|                                                      | First-dose Vaccination Rate |                     |
| SSIV                                                 | 0.904***<br>(0.129)         | 0.902***<br>(0.129) |
| Proportion of Population being White                 |                             | 25.067<br>(31.790)  |
| Proportion of Population without High School Degrees |                             | -35.482<br>(38.266) |
| Poverty Rate                                         |                             | -8.274<br>(22.299)  |
| Num.Obs.                                             | 1 673 351                   | 1 673 343           |
| FE: date                                             | X                           | X                   |
| FE: county                                           | X                           | X                   |

**Table A9: First-stage Results**

Note: This table presents the first-stage relationship between the SSIV and the first-dose vaccination rate. Columns (1)-(2) present the coefficients without and with covariates, respectively. Covariates include the percentage of the county population being white, the percentage of the population without high school degrees, and the poverty rate. County and day fixed effects are included in all columns. Standard errors clustered at the county level are included in the parentheses. The estimation period is from December 13, 2020 to June 16, 2022.

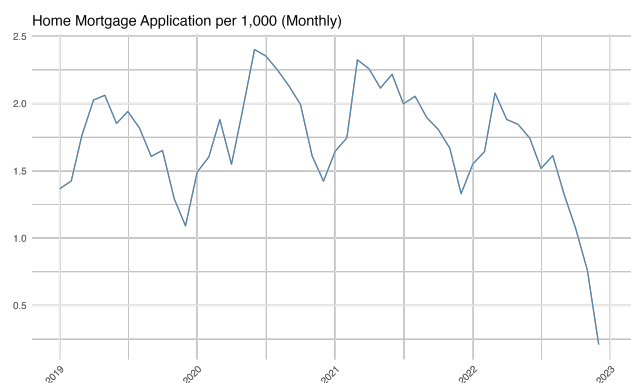
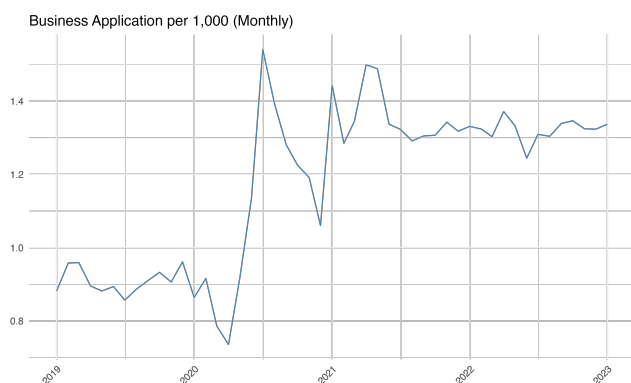
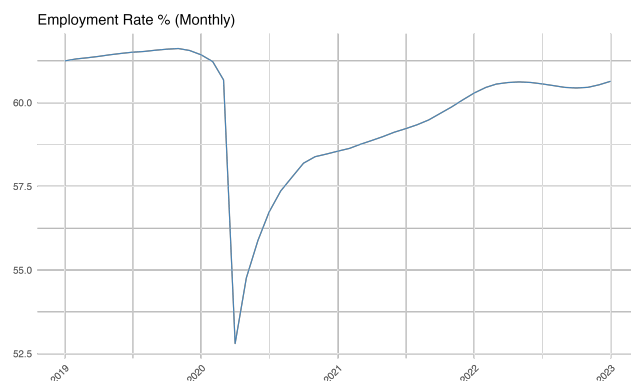
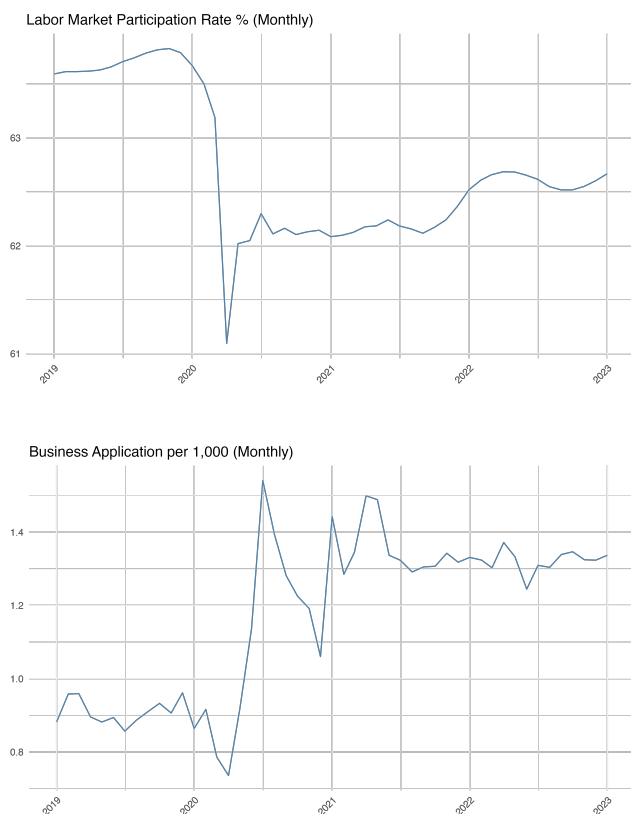


Figure A3: Trends for Four Economic Indicators

Note: This figure presents the national trends of the four economic indicators analyzed in the main section. From the top-left, clockwise: labor force participation rate, employment rate, home mortgage applications per 1,000, and business applications per 1,000.